
Chapter Three

Solution of Nonlinear Equations

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3.0 Introduction

This chapter is devoted to the problem of locating **roots** of equations (or **zeros** of functions).

The problem occurs frequently in scientific work. In this chapter, we are interested in *solving* a **nonlinear** equation or a system of nonlinear equations—namely, finding x such that $f(x) = 0$ or finding $X = (x_1, x_2, \dots, x_n)^T$ so that $F(X) = 0$. In these equations, one or more of the variables appear in any number of nonlinear ways. In contrast to this, we discuss finding the solution of linear systems of the form $Ax = b$ in Chapter 4 (p. 139).

The general problem, posed in the simplest case of a real-valued function of a real variable, is this: Given a function $f: \mathbb{R} \rightarrow \mathbb{R}$, find the values of x for which $f(x) = 0$. We shall consider several standard procedures for solving this problem.

Examples of nonlinear equations can be found in many applications. In the theory of diffraction of light, we need the roots of the equation

$$x - \tan x = 0$$

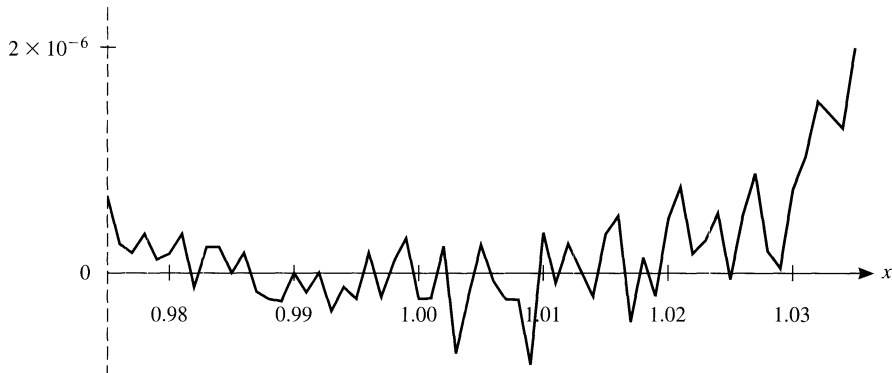
In the calculation of planetary orbits, we need the roots of **Kepler's equation**

$$x - a \sin x = b$$

for various values of a and b .

Since locating the zeros of functions has been an active area of study for several hundred years, numerous methods have been developed. In this chapter, we begin with three simple methods that are quite useful: the bisection method, Newton's method, and the secant method. Then we explore the general theory of fixed-point

FIGURE 3.1
Plot of expanded
polynomial



methods and continuation methods. Also, we discuss special methods for computing the zeros of polynomials.

When a computer is used to find an **approximate zero** of a function, there may be many approximate solutions even though the exact solution is unique. This is nicely illustrated by considering the polynomial

$$p_4(x) = x^4 - 4x^3 + 6x^2 - 4x + 1$$

If we happen to notice that this polynomial can be factored so that $p_4(x) = (x - 1)^4$, then it is obvious that 1 is the only zero (of multiplicity four). Suppose that we are given the expanded polynomial and do not see that it has such a zero. If we write a single-precision program in a computer, such as the `MarC-32`, and evaluate the expanded polynomial at points 0.001 distance apart in the interval $[0.975, 1.035]$, we find a large number of sign changes indicating the presence of apparent zeros. Plotting these points connected by straight lines, we obtain the graph shown in Figure 3.1. Instead of a nice clean curve for the polynomial, we obtain a **fuzzy** one. Any value in the interval $[0.981, 1.026]$ could be taken as an approximation to the true solution. The reasons for this are related to the way the polynomial was evaluated, the limited precision of the computer arithmetic being used, and the associated roundoff errors. This example follows a similar one found in Conte and de Boor [1980, p. 73], and it illustrates one of the dangers involved in finding roots.

3.1 Bisection (Interval Halving) Method

If f is a continuous function on the interval $[a, b]$ and if $f(a)f(b) < 0$, then f must have a zero in (a, b) . Since $f(a)f(b) < 0$, the function f changes sign on the interval $[a, b]$ and, therefore, it has at least one zero in the interval. This is a consequence of the Intermediate-Value Theorem (p. 5).

The bisection method exploits this idea in the following way. If $f(a)f(b) < 0$, then we compute $c = \frac{1}{2}(a + b)$ and test whether $f(a)f(c) < 0$. If this is true, then f

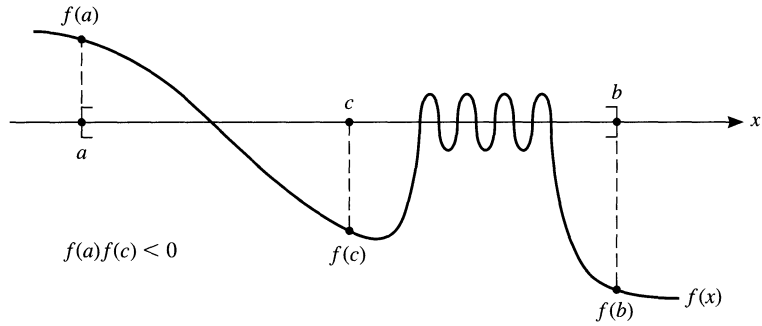


FIGURE 3.2(a)
Bisection method selects
left subinterval

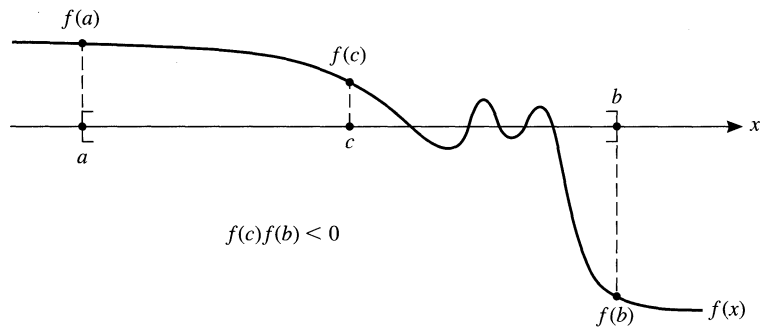


FIGURE 3.2(b)
Bisection method selects
right subinterval

has a zero in $[a, c]$. So we rename c as b and start again with the new interval $[a, b]$, which is half as large as the original interval. If $f(a)f(c) > 0$, then $f(c)f(b) < 0$, and in this case we rename c as a . In either case, a new interval containing a zero of f has been produced, and the process can be repeated. Figures 3.2(a) and (b) show the two cases, assuming $f(a) > 0 > f(b)$. These figures make it clear why the bisection method finds one zero but not all the zeros in the interval $[a, b]$. Of course, if $f(a)f(c) = 0$, then $f(c) = 0$ and a zero has been found. However, it is quite unlikely that $f(c)$ is exactly 0 in the computer because of roundoff errors. Thus, the **stopping criterion** should *not* be whether $f(c) = 0$. A reasonable tolerance must be allowed, such as $|f(c)| < 10^{-5}$ on the Marc-32. (See Section 2.1, p. 40, for a description of the hypothetical Marc-32 computer.) The bisection method is also known as the **method of interval halving**.

EXAMPLE 1 Use the bisection method to find the root of the equation $e^x = \sin x$ closest to 0.

Solution If the graphs of e^x and $\sin x$ are roughly plotted, it becomes clear that there are no positive roots of $f(x) = e^x - \sin x$ and that the first root to the left of 0 is in the

interval $[-4, -3]$. When the bisection algorithm was run on a machine similar to the Marc-32, the following output was produced, starting with the interval $[-4, -3]$:

k	c	$f(c)$
1	-3.5000	-0.321
2	-3.2500	-0.694×10^{-1}
3	-3.1250	0.605×10^{-1}
4	-3.1875	0.625×10^{-1}
$\vdots$	$\vdots$	$\vdots$
13	-3.1829	0.122×10^{-3}
14	-3.1830	0.193×10^{-4}
15	-3.1831	-0.124×10^{-4}
16	-3.1831	0.345×10^{-5}

Bisection Algorithm

Some parts of the bisection pseudocode (given on pp. 76–78) need some additional explanation. First, the midpoint c is computed as $c \leftarrow a + (b - a)/2$ rather than as $c \leftarrow (a + b)/2$ to adhere to the general stratagem that in numerical calculations it is best to compute a quantity by adding a small correction term to a previous approximation. Forsythe, Malcolm, and Moler [1977, p. 162] present an example in which the midpoint computed by $(a + b)/2$ moves outside of the interval $[a, b]$ on a machine with limited precision! Second, it is better to determine whether the function changes sign over the interval using $\text{sign}(w) \neq \text{sign}(u)$ rather than $wv < 0$, since the latter requires an unnecessary multiplication and could cause an underflow or overflow. Next, e is the computation of the error bound that is established in the theorem to follow. Finally, notice that *three* stopping criteria are present in the algorithm. First, M gives the maximum number of steps that the user permits. Such a safeguard should *always* be present to reduce the possibility of the computation going into an infinite loop. Next, the calculation can be stopped when either the error is small enough or the value of $f(c)$ is small enough. The parameters δ and ε control this. Examples can be easily given in which one of the latter two stopping criteria is satisfied but the other is not. For example, consider the two sketches in Figures 3.3(a) and (b). In Figure 3.3(a), the graph is flat near the zero, which corresponds to a multiple root, and the bisection method may have difficulty determining this zero to high precision. The curve in Figure 3.3(b) is, of course, not continuous, but the continuity of a function may be hard to verify beforehand. Although these are pathological examples, we choose to stop the algorithm when any one of the three stopping criteria is satisfied, in the interest of having a robust code.

The bisection algorithm can be written as follows:

```
input  $a, b, M, \delta, \varepsilon$ 
 $u \leftarrow f(a)$ 
 $v \leftarrow f(b)$ 
 $e \leftarrow b - a$ 
output  $a, b, u, v$ 
```

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FIGURE 3.3(a)
Criterion $|b - a| < \delta$
failure

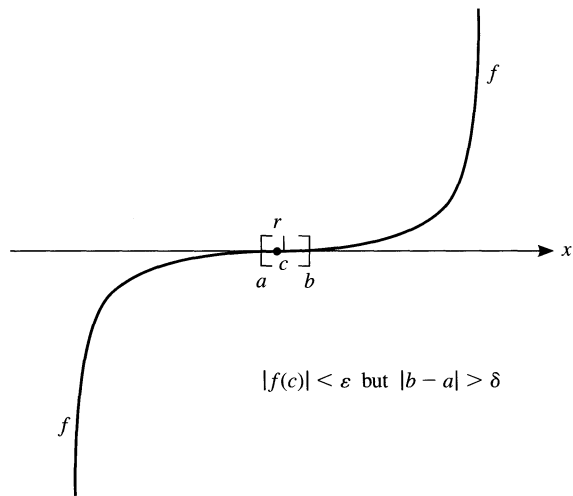
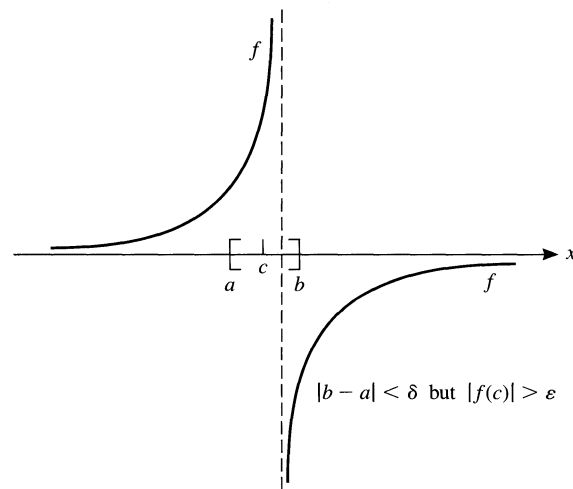


FIGURE 3.3(b)
Criterion $|f(c)| < \varepsilon$
failure



```
if sign(u) = sign(v) then stop
for k = 1 to M do
  e ← e/2
  c ← a + e
  w ← f(c)
  output k, c, w, e
  if |e| < δ or |w| < ε then stop
  if sign(w) ≠ sign(u) then
```

```

        b ← c
        v ← w
    else
        a ← c
        u ← w
    end if
end do

```

Error Analysis

To analyze the bisection method, let us denote the successive intervals that arise in the process by $[a_0, b_0]$, $[a_1, b_1]$, and so on. Here are some observations about these numbers:

$$\begin{aligned}
 a_0 &\leq a_1 \leq a_2 \leq \cdots \leq b_0 \\
 b_0 &\geq b_1 \geq b_2 \geq \cdots \geq a_0 \\
 b_{n+1} - a_{n+1} &= \frac{1}{2}(b_n - a_n) \quad (n \geq 0)
 \end{aligned} \tag{1}$$

Since the sequence $[a_n]$ is nondecreasing and bounded above, it converges. Likewise, $[b_n]$ converges. If we apply Equation (1) repeatedly, we find that

$$b_n - a_n = 2^{-n}(b_0 - a_0)$$

Thus,

$$\lim_{n \rightarrow \infty} b_n - \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} 2^{-n}(b_0 - a_0) = 0$$

If we put

$$r = \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n$$

then, by taking a limit in the inequality $0 \geq f(a_n)f(b_n)$, we obtain $0 \geq [f(r)]^2$, whence $f(r) = 0$.

Suppose that, at a certain stage in the process, the interval $[a_n, b_n]$ has just been defined. If the process is now stopped, the root is certain to lie in this interval. The best estimate of the root at this stage is not a_n or b_n but the **midpoint** of the interval:

$$c_n = (a_n + b_n)/2$$

The error is then bounded as follows:

$$|r - c_n| \leq \frac{1}{2}(b_n - a_n) = 2^{-(n+1)}(b_0 - a_0)$$

Summarizing this discussion, we have the following theorem on the bisection method.

THEOREM 1 Theorem on Bisection Method

If $[a_0, b_0], [a_1, b_1], \dots, [a_n, b_n], \dots$ denote the intervals in the bisection method, then the limits $\lim_{n \rightarrow \infty} a_n$ and $\lim_{n \rightarrow \infty} b_n$ exist, are equal, and represent a zero of f . If $r = \lim_{n \rightarrow \infty} c_n$ and $c_n = \frac{1}{2}(a_n + b_n)$, then

$$|r - c_n| \leq 2^{-(n+1)}(b_0 - a_0) \quad (2)$$

EXAMPLE 2 Suppose that the bisection method is started with the interval $[50, 63]$. How many steps should be taken to compute a root with relative accuracy of one part in 10^{-12} ?

Solution The stated requirement on relative accuracy means that

$$|r - c_n|/|r| \leq 10^{-12}$$

We know that $r \geq 50$, and thus it suffices to secure the inequality

$$|r - c_n|/50 \leq 10^{-12}$$

By means of Theorem 1, we infer that the following condition is sufficient:

$$2^{-(n+1)} \times (13/50) \leq 10^{-12}$$

Solving this for n , we conclude that $n \geq 37$. ■

PROBLEMS 3.1

1. Find a positive root of

$$x^2 - 4x \sin x + (2 \sin x)^2 = 0$$

accurate to two significant figures. Use a hand calculator.

2. Consider the bisection method starting with the interval $[1.5, 3.5]$.

- What is the width of the interval at the n th step of this method?
- What is the maximum distance possible between the root r and the midpoint of this interval?

3. If the bisection method is used (in single precision) on the `Marc-32` starting with the interval $[128, 129]$, can we compute the root with absolute accuracy $< 10^{-6}$?

4. Determine the formula

$$n \geq \frac{\log(b_0 - a_0) - \log \varepsilon}{\log 2} - 1$$

involving $b_0 - a_0$ and ε for the number of steps that must be taken in the bisection method to guarantee that $|r - c_n| \leq \varepsilon$.

5. Determine the formula

$$n \geq \frac{\log(b_0 - a_0) - \log \varepsilon - \log a_0}{\log 2} - 1$$

involving a_0 , b_0 , and ε for the number of steps that should be taken in the bisection algorithm to ensure that the root is determined with *relative* accuracy $\leq \varepsilon$. Assume $a_0 > 0$.

6. (Continuation) What happens in the preceding problem if $a_0 < 0 < b_0$?
7. If the bisection method is used starting with the interval $[2, 3]$, how many steps must be taken to compute a root with absolute accuracy $< 10^{-6}$? Answer the same question for the *relative* accuracy. What about to full single precision on the `MarC-32` in each case?
8. Let $c_n = \frac{1}{2}(a_n + b_n)$, $r = \lim_{n \rightarrow \infty} c_n$, and $e_n = r - c_n$. Here $[a_n, b_n]$, with $n \geq 0$, denotes the successive intervals that arise in the bisection method when it is applied to a continuous function f .
- a. Show that $|e_n| \leq 2^{-n}(b_1 - a_1)$.
 - b. Show that $e_n = \mathcal{O}(2^{-n})$ as $n \rightarrow \infty$.
 - c. Is it true that $|e_0| \geq |e_1| \geq \dots$? Explain.
 - d. Show that $|c_n - c_{n+1}| = 2^{-n-2}(b_0 - a_0)$.
 - e. Show that for all n and m , $a_m \leq b_n$.
 - f. Show that r is the unique element in $\bigcap_{n=0}^{\infty} [a_n, b_n]$.
 - g. Show that for all n , $[a_n, b_n] \supset [a_{n+1}, b_{n+1}]$.
9. In the bisection method, an interval $[a_{n-1}, b_{n-1}]$ is divided in half, and one of these halves is chosen for the next interval. Define $d_n = 0$ if $[a_n, b_n]$ is the left half of the interval $[a_{n-1}, b_{n-1}]$, and let $d_n = 1$ otherwise. Express the root determined by the algorithm in terms of the sequence $d_1, d_2, \dots$. *Hint:* Consider the case $[a_0, b_0] = [0, 1]$ first, and think about the binary representation of the root.
10. Using the notation of the previous two problems, find the formulas that relate a_n, b_n , and c_n to d_n .
11. Give an example (or prove that none exists) in which $a_0 < a_1 < a_2 < \dots$.
12. Give an example in which $a_0 = a_1 < a_2 = a_3 < a_4 = a_5 < a_6 = \dots$.
13. In the bisection method, does $\lim_{n \rightarrow \infty} |r - c_{n+1}|/|r - c_n|$ exist? Explain.
14. Let the bisection method be applied to a continuous function, resulting in intervals $[a_0, b_0]$, $[a_1, b_1]$, and so on. Let $r = \lim_{n \rightarrow \infty} a_n$. Which of these statements can be false?
- a. $a_0 \leq a_1 \leq a_2 \leq \dots$
 - b. $|r - 2^{-1}(a_n + b_n)| \leq 2^{-n}(b_0 - a_0) \quad (n \geq 0)$
 - c. $|r - 2^{-1}(a_{n+1} + b_{n+1})| \leq |r - 2^{-1}(a_n + b_n)| \quad (n \geq 0)$
 - d. $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n] \quad (n \geq 0)$
 - e. $|r - a_n| = \mathcal{O}(2^{-n})$ as $n \rightarrow \infty$
 - f. $|r - c_n| < |r - c_{n-1}| \quad (n \geq 1)$
15. Prove that the point c computed in the bisection method is the point where the line through $(a, \text{sign}(f(a)))$ and $(b, \text{sign}(f(b)))$ intersects the x -axis.
16. Suppose that $|a_n - b_n| \leq \lambda_n |a_{n-1} - b_{n-1}|$ for all n with $\lambda_n < 1$. Find an upper bound on $|a_n - b_n|$ in terms of $|a_0 - b_0|$ and $\lambda = \max_{1 \leq i \leq n} \{\lambda_i\}$.

COMPUTER
PROBLEMS 3.1

1. Write and test a subprogram or procedure to implement the bisection algorithm. Test the program on these functions and intervals:
- a. $x^{-1} - \tan x$ on $[0, \pi/2]$
 - b. $x^{-1} - 2^x$ on $[0, 1]$

- c. $2^{-x} + e^x + 2 \cos x - 6$ on $[1, 3]$
- d. $(x^3 + 4x^2 + 3x + 5)/(2x^3 - 9x^2 + 18x - 2)$ on $[0, 4]$
- 2. Find numbers $a < b$ such that the two mathematically equivalent computations $c \leftarrow (a + b)/2$ and $c \leftarrow a + 0.5(b - a)$ produce different results on your computer. Do not choose an example in which there is underflow or overflow.
- 3. Find a root of $f(x) = x - \tan x$ in the interval $[1, 2]$.
- 4. Find a root of

$$x^8 - 36x^7 + 546x^6 - 4536x^5 + 22449x^4 - 67284x^3 + 118124x^2 - 109584x + 40320 = 0$$

in the interval $[5.5, 6.5]$. Change -36 to -36.001 and repeat.

- 5. Write and test a recursive version of the bisection algorithm.

3.2 Newton's Method

Newton's method is a general procedure that can be applied in many diverse situations. When specialized to the problem of locating a zero of a real-valued function of a real variable, it is often called the **Newton-Raphson iteration**. In general, Newton's method is faster than the bisection and the secant methods since its convergence is quadratic rather than linear or superlinear. Once the quadratic convergence becomes effective, that is, the values of Newton's method sequence are sufficiently close to the root, the convergence is so rapid that only a few more values are needed. Unfortunately, the method is not guaranteed always to converge. Often Newton's method is combined with other slower methods in a hybrid method that is numerically globally convergent.

As in Section 3.1 (p. 74), we have a function f whose zeros are to be determined numerically. Let r be a zero of f and let x be an approximation to r . If f'' exists and is continuous, then by Taylor's Theorem,

$$0 = f(r) = f(x + h) = f(x) + hf'(x) + \mathcal{O}(h^2)$$

where $h = r - x$. If h is small (that is, x is near r), then it is reasonable to ignore the $\mathcal{O}(h^2)$ term and solve the remaining equation for h . If we do this, the result is $h = -f(x)/f'(x)$. If x is an approximation to r , then $x - f(x)/f'(x)$ should be a better approximation to r . Newton's method begins with an estimate x_0 of r and then defines inductively

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \qquad (n \geq 0) \tag{1}$$

Newton's Algorithm

A simple algorithm that applies M steps of Newton's method, starting with an initial value for x , is:

```
input x, M
y ← f(x)
output 0, x, y
for k = 1 to M do
    x ← x - y/f'(x)
    y ← f(x)
    output k, x, y
end do
```

A more detailed pseudocode including stopping criteria follows:

```
input x0, M, δ, ε
v ← f(x0)
output 0, x0, v
if |v| < ε then stop
for k = 1 to M do
    x1 ← x0 - v/f'(x0)
    v ← f(x1)
    output k, x1, v
    if |x1 - x0| < δ or |v| < ε then stop
    x0 ← x1
end do
```

A computer program based on either of these two algorithms would need subprograms or procedures for $f(x)$ and $f'(x)$.

EXAMPLE 1 Use Newton’s method, with double-precision computation, to find the negative zero of the function $f(x) = e^x - 1.5 - \tan^{-1} x$.

Solution The preceding algorithm was executed with double precision on a machine with 48-bit mantissas in single precision. (Double-precision machine numbers have 96 bits, corresponding to about 28 decimal places.) The function $f'(x) = e^x - (1 + x^2)^{-1}$, as well as f , had to be programmed. A starting point of $x_0 = -7$ was chosen. The output from the computer program is shown here.

k	x	$f(x)$
0	- 7.00000 00000 00000 00000 0	-0.702×10^{-1}
1	-10.67709 61766 40013 99296 98438 6	-0.226×10^{-1}
2	-13.27916 73756 32712 90859 78631 9	-0.437×10^{-2}
3	-14.05365 58542 69238 73474 83175 3	-0.239×10^{-3}
4	-14.10110 99568 66413 47616 31270 6	-0.800×10^{-6}
5	-14.10126 97709 39415 94621 57950 6	-0.901×10^{-11}
6	-14.10126 97727 39968 42508 30031 4	-0.114×10^{-20}
7	-14.10126 97727 39968 42531 15512 2	0.000
8	-14.10126 97727 39968 42531 15512 2	0.000

The output shows rapid convergence of the iterates; in fact, in each step the number of correct digits in the approximation seems to *double*. Our analysis will reveal why this is true. ■

Graphical Interpretation

Before examining the theoretical basis for Newton’s method, let’s give a graphical interpretation of it. From the description already given, we can say that Newton’s method involves **linearizing the function**. That is, f was replaced by a linear function. The usual way of doing this is to replace f by the first two terms in its Taylor series. Thus, if

$$f(x) = f(c) + f'(c)(x - c) + \frac{1}{2!}f''(c)(x - c)^2 + \cdots$$

then the linearization (at c) produces the linear function

$$\ell(x) = f(c) + f'(c)(x - c)$$

Notice that ℓ is a good approximation to f in the vicinity of c , and in fact we have $\ell(c) = f(c)$ and $\ell'(c) = f'(c)$. Thus, the linear function has the same value and the same slope as f at the point c . So in Newton’s method we are constructing the tangent to the f -curve at a point near r , and finding where the tangent line intersects the x -axis. (See Figure 3.4.)

Keeping in mind this graphical interpretation, we can easily imagine functions and starting points for which the Newton iteration *fails*. Such a function is shown in Figure 3.5. In this example, the shape of the curve is such that for certain starting values, the sequence $[x_n]$ diverges. Thus, any formal statement about Newton’s

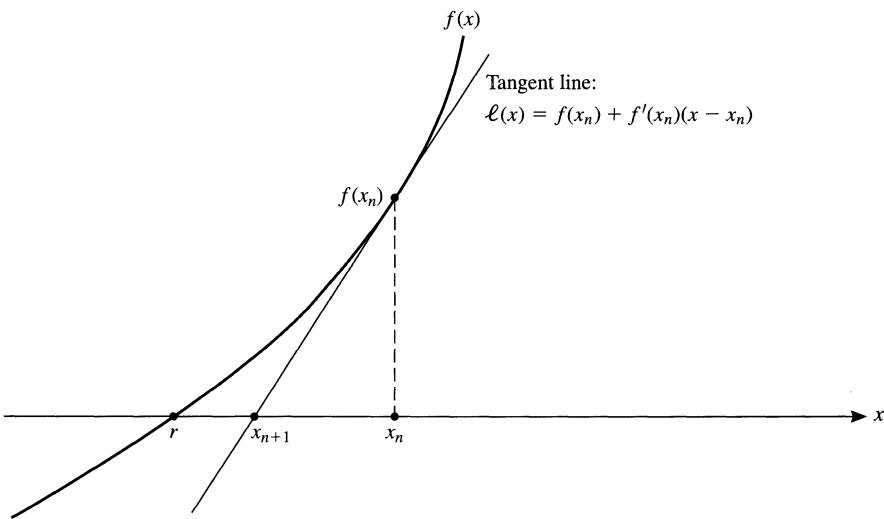
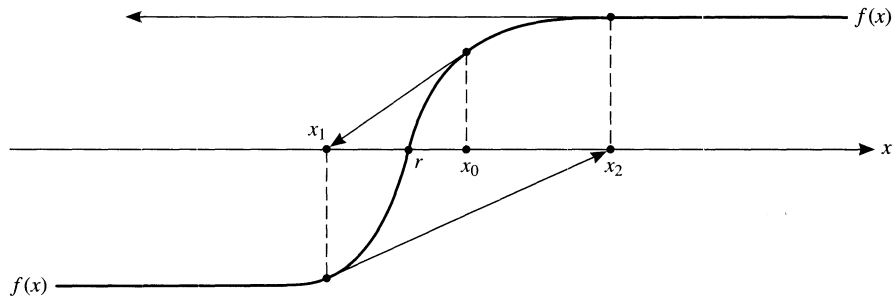


FIGURE 3.4
Geometric interpretation
of Newton’s method

FIGURE 3.5
Example of
nonconvergence of
Newton's method



method must involve an assumption that x_0 is **sufficiently close to a zero** or that the graph of f has a prescribed shape.

Error Analysis

Now we shall analyze the errors in Newton's method. By **errors**, we mean the quantities

$$e_n = x_n - r$$

(We are *not* considering roundoff errors.) Let us assume that f'' is continuous and r is a **simple zero** of f , so that $f(r) = 0 \neq f'(r)$. From the definition of the Newton iteration, we have

$$\begin{aligned} e_{n+1} &= x_{n+1} - r = x_n - \frac{f(x_n)}{f'(x_n)} - r \\ &= e_n - \frac{f(x_n)}{f'(x_n)} = \frac{e_n f'(x_n) - f(x_n)}{f'(x_n)} \end{aligned} \tag{2}$$

By Taylor's Theorem, we have

$$0 = f(r) = f(x_n - e_n) = f(x_n) - e_n f'(x_n) + \frac{1}{2} e_n^2 f''(\xi_n)$$

where ξ_n is a number between x_n and r . A rearrangement of this equation yields

$$e_n f'(x_n) - f(x_n) = \frac{1}{2} f''(\xi_n) e_n^2$$

Putting this in Equation (2) leads to

$$e_{n+1} = \frac{1}{2} \frac{f''(\xi_n)}{f'(x_n)} e_n^2 \approx \frac{1}{2} \frac{f''(r)}{f'(r)} e_n^2 = C e_n^2 \tag{3}$$

Suppose that $C \approx 1$ and $e_n \approx 10^{-4}$. Then by Equation (3), we have $e_{n+1} \approx 10^{-8}$ and $e_{n+2} \approx 10^{-16}$. We are impressed that only a few additional iterations are needed to obtain more than machine precision!

This equation tells us that e_{n+1} is roughly a constant times e_n^2 . This desirable state of affairs is called **quadratic convergence**. It accounts for the apparent doubling of precision with each iteration of Newton’s method in many applications.

We have yet to establish the convergence of the method. By Equation (3), the idea of the proof is simple: If e_n is small and if the factor $\frac{1}{2}f''(\xi_n)/f'(x_n)$ is not too large, then e_{n+1} will be smaller than e_n . Define a quantity $c(\delta)$ dependent on δ by

$$c(\delta) = \frac{1}{2} \max_{|x-r| \leq \delta} |f''(x)| \bigg/ \min_{|x-r| \leq \delta} |f'(x)| \quad (\delta > 0) \tag{4}$$

We select δ small enough to be sure that the denominator in Equation (4) is positive, and then if necessary we decrease δ so that $\delta c(\delta) < 1$. This is possible because as δ converges to 0, $c(\delta)$ converges to $\frac{1}{2}|f''(r)|/|f'(r)|$, and so $\delta c(\delta)$ converges to 0. Having fixed δ , set $\rho = \delta c(\delta)$. Suppose that we start the Newton iteration with a point x_0 satisfying $|x_0 - r| \leq \delta$. Then $|e_0| \leq \delta$ and $|\xi_0 - r| \leq \delta$. Hence, by the definition of $c(\delta)$, we have

$$\frac{1}{2} |f''(\xi_0)/f'(x_0)| \leq c(\delta)$$

Therefore, Equation (3) yields

$$|x_1 - r| = |e_1| \leq e_0^2 c(\delta) = |e_0| |e_0| c(\delta) \leq |e_0| \delta c(\delta) = |e_0| \rho < |e_0| \leq \delta$$

This shows that the next point, x_1 , also lies within δ units of r . Hence, the argument can be repeated, with the results

$$\begin{aligned} |e_1| &\leq \rho |e_0| \\ |e_2| &\leq \rho |e_1| \leq \rho^2 |e_0| \\ |e_3| &\leq \rho |e_2| \leq \rho^3 |e_0| \\ &\vdots \end{aligned}$$

In general, we have

$$|e_n| \leq \rho^n |e_0|$$

Since $0 \leq \rho < 1$, we have $\lim_{n \rightarrow \infty} \rho^n = 0$, and so $\lim_{n \rightarrow \infty} e_n = 0$. Summarizing, we obtain the following theorem on Newton’s method.

THEOREM 1 Theorem on Newton’s Method

Let f'' be continuous and let r be a simple zero of f . Then there is a neighborhood of r and a constant C such that if Newton’s method is started in that neighborhood, the successive points become steadily closer to r and satisfy

$$|x_{n+1} - r| \leq C(x_n - r)^2 \quad (n \geq 0)$$

In some situations Newton's iteration can be guaranteed to converge from an *arbitrary* starting point. We give one such theorem as a sample.

THEOREM 2 Theorem on Newton's Method for a Convex Function

If f belongs to $C^2(\mathbb{R})$, is increasing, is convex, and has a zero, then the zero is unique, and the Newton iteration will converge to it from any starting point.

Proof Recall that a function f is **convex** if $f''(x) > 0$ for all x . Since f is increasing, $f' > 0$ on $\mathbb{R}$. By Equation (3), $e_{n+1} > 0$. Thus, $x_n > r$ for $n \geq 1$. Since f is increasing, $f(x_n) > f(r) = 0$. Hence, by Equation (2), $e_{n+1} < e_n$. Thus, the sequences $[e_n]$ and $[x_n]$ are decreasing and bounded below (by 0 and r , respectively). Therefore, the limits $e^* = \lim_{n \rightarrow \infty} e_n$ and $x^* = \lim_{n \rightarrow \infty} x_n$ exist. From Equation (2), we have $e^* = e^* - f(x^*)/f'(x^*)$, whence $f(x^*) = 0$ and $x^* = r$. ■

EXAMPLE 2 Find an efficient method for computing square roots using Newton's method.

Solution Let $R > 0$ and set $x = \sqrt{R}$. Then x is a root of the equation $x^2 - R = 0$. If we use Newton's method (1) on the function $f(x) = x^2 - R$, the iteration formula can be written as

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{R}{x_n} \right)$$

This formula, combined with range reduction, is often used in the subroutine that computes square roots. (This formula is very ancient and is credited to Heron, a Greek engineer and architect who lived sometime between 100 B.C. and A.D. 100.) If, for example, we wish to compute $\sqrt{17}$ and begin with $x_0 = 4$, the successive approximations are as follows (given in rounded form to exhibit only correct figures):

$$\begin{aligned} x_1 &= 4.12 \\ x_2 &= 4.123106 \\ x_3 &= 4.1231056256177 \\ x_4 &= 4.123105625617660549821409856 \end{aligned}$$

The value given by x_4 is correct to 28 figures, and we observe the expected doubling of significant digits in the results. ■

Implicit Functions

One interesting application of Newton's method occurs in the evaluation of functions defined implicitly. In Section 1.2 (p. 23), the Implicit Function Theorem was discussed; it states that under rather general conditions, the equation $G(x, y) = 0$ defines y as a function of x . If x is prescribed, then the equation $G(x, y) = 0$ can be solved for y using Newton's method. From a suitable starting point y_0 , we define $y_1, y_2, \dots$ by

$$y_{k+1} = y_k - G(x, y_k) \bigg/ \frac{\partial G}{\partial y}(x, y_k)$$

This method can be used to construct a table of the function $y(x)$. If the table contains an entry (x_n, y_n) , and if we desire to compute a nearby entry (x_{n+1}, y_{n+1}) , we *start* the Newton iteration with (x_{n+1}, y_n) . The *result* of the iteration will be the correct value y_{n+1} that makes the equation $G(x_{n+1}, y_{n+1}) = 0$ true. Since $G(x_n, y_n) = 0$ and x_{n+1} is close to x_n , we can expect that $G(x_{n+1}, y_n)$ is small and that a few steps of the Newton method will produce the correction to y_n necessary to achieve the exact equality $G(x_{n+1}, y_{n+1}) = 0$.

EXAMPLE 3 Produce a table of x versus y , where y is defined implicitly as a function of x . Use $G(x, y) = 3x^7 + 2y^5 - x^3 + y^3 - 3$ and start at $x = 0$, proceeding in steps of 0.1 to $x = 10$.

Solution Let $x = 0$ and $y = 1$. We assume that four steps of Newton's method will suffice to give full machine precision. In the algorithm, we have M steps in the x -variable, and at each step there are N iterations in Newton's method. The program based on the algorithm would need two subprograms or procedures, the first to compute $G(x, y)$ and the second to compute $\partial G/\partial y$. (The latter happens to be independent of x in this example.) These functions are

$$G(x, y) = 3x^7 + 2y^5 - x^3 + y^3 - 3$$
$$\frac{\partial G}{\partial y}(x, y) = 10y^4 + 3y^2$$

Here is a suitable algorithm:

```
input x ← 0; y ← 1; h ← 0.1; M ← 100; N ← 4
output 0, x, y, G(x, y)
for i = 1 to M do
  x ← x + h
  for j = 1 to N do
    y ← y - G(x, y) / (∂G/∂y)(x, y)
  end do
  output i, x, y, G(x, y)
end do
```

Some of the values of the implicit function, computed by the foregoing algorithm, are shown here:

i	x	y	$G(x, y)$
0	0.0	1.00000 0	0.00
1	0.1	1.00007 7	0.00
2	0.2	1.00061 2	0.89×10^{-15}
⋮	⋮	⋮	⋮
20	2.0	- 2.81063 9	-0.82×10^{-10}
⋮	⋮	⋮	⋮
80	8.0	-19.9263 5	0.56×10^{-9}

$$\begin{array}{cccc} \vdots & \vdots & \vdots & \vdots \\ 99 & 9.9 & -26.85618 & 0.12 \times 10^{-7} \\ 100 & 10.0 & -27.23685 & -0.15 \times 10^{-8} \end{array}$$

■

Systems of Nonlinear Equations

Newton’s method for systems of nonlinear equations follows the same strategy that was used for a single equation. Thus, we **linearize** and **solve**, repeating the steps as often as necessary. Let us illustrate with a pair of equations involving two variables:

$$\begin{cases} f_1(x_1, x_2) = 0 \\ f_2(x_1, x_2) = 0 \end{cases} \tag{5}$$

Supposing that (x_1, x_2) is an approximate solution of (5), let us compute corrections h_1 and h_2 so that $(x_1 + h_1, x_2 + h_2)$ will be a better approximate solution. Using only linear terms in the Taylor expansion in two variables (see Section 1.1, p. 11), we have

$$\begin{cases} 0 = f_1(x_1 + h_1, x_2 + h_2) \approx f_1(x_1, x_2) + h_1 \frac{\partial f_1}{\partial x_1} + h_2 \frac{\partial f_1}{\partial x_2} \\ 0 = f_2(x_1 + h_1, x_2 + h_2) \approx f_2(x_1, x_2) + h_1 \frac{\partial f_2}{\partial x_1} + h_2 \frac{\partial f_2}{\partial x_2} \end{cases} \tag{6}$$

The partial derivatives appearing in (6) are to be evaluated at (x_1, x_2) . Equation (6) constitutes a pair of *linear* equations for determining h_1 and h_2 . The coefficient matrix is the **Jacobian matrix** of f_1 and f_2 :

$$J = \begin{bmatrix} \partial f_1 / \partial x_1 & \partial f_1 / \partial x_2 \\ \partial f_2 / \partial x_1 & \partial f_2 / \partial x_2 \end{bmatrix}$$

To solve (6), we require J to be nonsingular. If this condition is met, the solution is

$$\begin{bmatrix} h_1 \\ h_2 \end{bmatrix} = -J^{-1} \begin{bmatrix} f_1(x_1, x_2) \\ f_2(x_1, x_2) \end{bmatrix}$$

Hence, Newton’s method for two nonlinear equations in two variables is

$$\begin{bmatrix} x_1^{(k+1)} \\ x_2^{(k+1)} \end{bmatrix} = \begin{bmatrix} x_1^{(k)} \\ x_2^{(k)} \end{bmatrix} + \begin{bmatrix} h_1^{(k)} \\ h_2^{(k)} \end{bmatrix}$$

where the Jacobian linear system

$$J \begin{bmatrix} h_1^{(k)} \\ h_2^{(k)} \end{bmatrix} = - \begin{bmatrix} f_1(x_1^{(k)}, x_2^{(k)}) \\ f_2(x_1^{(k)}, x_2^{(k)}) \end{bmatrix}$$

is solved using Gaussian elimination—a subject discussed in Chapter 4 (p. 139). Solving the Jacobian system may be difficult if J is nearly singular.

To discuss larger systems involving many more variables, no new ideas are needed. However, a matrix-vector formalism is very convenient in this situation. The system of equations

$$f_i(x_1, x_2, \dots, x_n) = 0 \quad (1 \leq i \leq n) \quad (7)$$

can be expressed simply as

$$F(X) = 0 \quad (8)$$

by letting $X = (x_1, x_2, \dots, x_n)^T$ and $F = (f_1, f_2, \dots, f_n)^T$. The analogue of Equation (6) is

$$0 = F(X + H) \approx F(X) + F'(X)H \quad (9)$$

where $H = (h_1, h_2, \dots, h_n)^T$ and $F'(X)$ is the $n \times n$ Jacobian matrix $J(X)$ with elements $\partial f_i / \partial x_j$; namely,

$$F'(X) = \begin{bmatrix} \partial f_1 / \partial x_1 & \partial f_1 / \partial x_2 & \cdots & \partial f_1 / \partial x_n \\ \partial f_2 / \partial x_1 & \partial f_2 / \partial x_2 & \cdots & \partial f_2 / \partial x_n \\ \vdots & \vdots & \ddots & \vdots \\ \partial f_n / \partial x_1 & \partial f_n / \partial x_2 & \cdots & \partial f_n / \partial x_n \end{bmatrix}$$

The correction vector H is obtained by solving the linear system of equations in (9). Theoretically, this means

$$H = -F'(X)^{-1} F(X) \quad (10)$$

but in practice H would usually be determined by Gaussian elimination from (9), thus bypassing the more costly computation of the inverse in (10). Hence, Newton's method for n nonlinear equations in n variables is given by

$$X^{(k+1)} = X^{(k)} + H^{(k)} \quad (11)$$

where the Jacobian system is

$$F'(X^{(k)})H^{(k)} = -F(X^{(k)}) \quad (12)$$

EXAMPLE 4 Starting with $(1, 1, 1)^T$, carry out six iterations of Newton's method for finding a root of the nonlinear system

$$\begin{cases} xy = z^2 + 1 \\ xyz + y^2 = x^2 + 2 \\ e^x + z = e^y + 3 \end{cases}$$

Solution Let

$$F(X) = \begin{bmatrix} f_1(x_1, x_2, x_3) \\ f_2(x_1, x_2, x_3) \\ f_3(x_1, x_2, x_3) \end{bmatrix} = \begin{bmatrix} x_1 x_2 - x_3^2 - 1 \\ x_1 x_2 x_3 - x_1^2 + x_2^2 - 2 \\ e^{x_1} - e^{x_2} + x_3 - 3 \end{bmatrix}$$

Taking partial derivatives, we get the Jacobian matrix

$$F'(X) = \begin{bmatrix} x_2 & x_1 & -2x_3 \\ x_2x_3 - 2x_1 & x_1x_3 + 2x_2 & x_1x_2 \\ e^{x_1} & -e^{x_2} & 1 \end{bmatrix}$$

Using the starting value $X^{(0)} = (1, 1, 1)^T$, we carry out the nonlinear Newton's method given in Equations (11) and (12) and obtain the following:

n	x_1	x_2	x_3
0	1.00000 00	1.00000 00	1.00000 00
1	2.18932 60	1.59847 51	1.39390 06
2	1.85058 96	1.44425 14	1.27822 40
3	1.78016 11	1.42443 59	1.23929 24
4	1.77767 47	1.42396 09	1.23747 38
5	1.77767 19	1.42396 05	1.23747 11
6	1.77767 19	1.42396 05	1.23747 11

The solution of systems such as (8) is often challenging. The standard reference on the subject is Ortega and Rheinboldt [1970]. See also Rheinboldt [1974], Ostrowski [1966], Byrne and Hall [1973], Schnabel and Frank [1984], Eaves, Gould, Peitgen, and Todd [1983], and Allgower, Glasshoff, and Peitgen [1981]. For a discussion of the convergence of Newton's method in higher dimensions, refer to Goldstein [1966] or Ortega and Rheinboldt [1970].

PROBLEMS 3.2

- 1. Find the smallest positive starting point for which Newton's method diverges when it is applied to $f(x) = \tan^{-1} x$.
- 2. Let Newton's method be used on $f(x) = x^2 - q$ (where $q > 0$). Show that if x_n has k correct digits after the decimal point, then x_{n+1} will have at least $2k - 1$ correct digits after the decimal point, provided that $q > 0.006$ and $k \geq 1$.
- 3. Prove that if Newton's method is used on a function f for which f'' is continuous and $f(r) = 0 \neq f'(r)$, then $\lim_{n \rightarrow \infty} e_{n+1}/e_n^{-2}$ exists and equals $f''(r)/[2f'(r)]$. How can this fact be used in a program to test whether convergence is quadratic?
- 4. (Steffensen's method) Consider the iteration formula

$$x_{n+1} = x_n - f(x_n)/g(x_n)$$

where

$$g(x) = [f(x + f(x)) - f(x)]/f(x)$$

Show that this is quadratically convergent, under suitable hypotheses.

- 5. What is the purpose of the following iteration formula?

$$x_{n+1} = 2x_n - x_n^2y$$

Identify it as the Newton iteration for a certain function.

6. To compute reciprocals without division, we can solve $x = 1/R$ by finding a zero of the function $f(x) = x^{-1} - R$. Write a short algorithm to find $1/R$ by Newton's method applied to f . Do not use division or exponentiation in your algorithm. For positive R , what starting points are suitable?
7. Define $x_0 = 0$ and $x_{n+1} = x_n - [(\tan x_n - 1)/\sec^2 x_n]$. What is $\lim_{n \rightarrow \infty} x_n$ in this example? Relate this to Newton's method.
8. Perform four iterations of Newton's method for the polynomial

$$p(x) = 4x^3 - 2x^2 + 3$$

starting with $x_0 = -1$. Use a hand calculator.

9. If Newton's method is used on $f(x) = x^3 - 2$ starting with $x_0 = 1$, what is x_2 ?
10. Devise a Newton iteration formula for computing $\sqrt[3]{R}$ where $R > 0$. Perform a graphical analysis of your function $f(x)$ to determine the starting values for which the iteration will converge.
11. Devise a Newton algorithm for computing the fifth root of any positive real number.
12. The function $f(x) = x^2 + 1$ has zeros in the complex plane at $x = \pm i$. Is there a *real* starting point for complex Newton's method such that the iterates converge to either of these zeros? What complex starting points will work?
13. If Newton's method is used with $f(x) = x^2 - 1$ and $x_0 = 10^{10}$, how many steps are required to obtain the root with accuracy 10^{-8} ? (Solve analytically, not experimentally.)
14. Suppose that r is a double zero of the function f . Thus, $f(r) = f'(r) = 0 \neq f''(r)$. Show that if f'' is continuous, then in Newton's method we shall have $e_{n+1} \approx \frac{1}{2}e_n$ (linear convergence).
15. Consider a variation of Newton's method in which only one derivative is needed; that is,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_0)}$$

Find C and s such that

$$e_{n+1} = Ce_n^s$$

16. Prove that Newton's iteration will diverge for these functions, no matter what (real) starting point is selected.
 - a. $f(x) = x^2 + 1$
 - b. $f(x) = 7x^4 + 3x^2 + \pi$
17. Which of these sequences converges quadratically?
 - a. $1/n^2$
 - b. $1/2^{2^n}$
 - c. $1/\sqrt{n}$
 - d. $1/e^n$
 - e. $1/n^n$
18. Find the conditions on α to ensure that the iteration

$$x_{n+1} = x_n - \alpha f(x_n)$$

will converge linearly to a zero of f if started near the zero.

19. Prove that if r is a zero of multiplicity k of the function f , then quadratic convergence in Newton's iteration will be restored by making this modification:

$$x_{n+1} = x_n - kf(x_n)/f'(x_n)$$

20. (Continuation) In the course of using Newton's method, how can a multiple zero be detected by examining the behavior of the points $(x_n, f(x_n))$?
21. **Halley's method** for solving the equation $f(x) = 0$ uses the iteration formula

$$x_{n+1} = x_n - \frac{f_n f'_n}{(f'_n)^2 - (f_n f''_n)/2}$$

where $f_n = f(x_n)$, and so on. Show that this formula results when Newton's iteration is applied to the function $f/\sqrt{f'}$.

22. Starting with $(0, 0, 1)$, carry out an iteration of Newton's method for nonlinear systems on

$$\begin{cases} xy - z^2 = 1 \\ xyz - x^2 + y^2 = 2 \\ e^x - e^y + z = 3 \end{cases}$$

Explain the results.

23. Perform two iterations of Newton's method on these systems.

- a. Starting with $(0, 1)$

$$\begin{cases} 4x_1^2 - x_2^2 = 0 \\ 4x_1x_2 - x_1 = 1 \end{cases}$$

- b. Starting with $(1, 1)$

$$\begin{cases} xy^2 + x^2y + x^4 = 3 \\ x^3y^5 - 2x^5y - x^2 = -2 \end{cases}$$

COMPUTER
PROBLEMS 3.2

1. Write a computer program to solve the equation $x = \tan x$ by means of Newton's method. Find the roots nearest 4.5 and 7.7.
2. (Continuation) Write and test a program to compute the first ten roots of the equation $\tan x = x$. (This is much more difficult than the preceding computer problem.) *Cultural note:* If $\lambda_1, \lambda_2, \dots$ are all the positive roots of this equation, then $\sum_{i=1}^\infty \lambda_i^{-2} = 1/10$. (*Amer. Math. Monthly*, Oct. 1986, p. 660.)
3. Find the positive minimum point of the function $f(x) = x^{-2} \tan x$ by computing the zeros of f' using Newton's method.
4. Write a brief computer program to solve the equation $x^3 + 3x = 5x^2 + 7$ by Newton's method. Take ten steps starting at $x_0 = 5$.
5. The equation $2x^4 + 24x^3 + 61x^2 - 16x + 1 = 0$ has two roots near 0.1. Determine them by means of Newton's method.
6. In the first example of this section, investigate the sensitivity of the root to perturbations in the constant term.
7. For the function $f(z) = z^4 - 1$, carry out the complex Newton's method using as starting values all grid points on the mesh with uniform spacing 0.1 in the circle $|z| < 2$ in the complex plane. Assign the same color to all grid points that engender sequences

convergent to the same zero. Display the resulting four-color mesh on a color graphics computer terminal or on a color plotter.

8. Program the Newton algorithm in complex arithmetic, and test it on these functions with the given starting points.
 - a. $f(z) = z^2 + 1$, $z = 3 + i$
 - b. $f(z) = z + \sin z - 3$, $z = 2 - i$
 - c. $f(z) = z^4 + z^2 + 2 + 3i$, $z = 1$
9. Program and test Steffensen's method (Problem 3.2.4, p. 90) using the test functions in Computer Problems 3.2.1–2 (p. 92).
10. The polynomial $p(x) = x^3 + 94x^2 - 389x + 294$ has zeros 1, 3, and -98 . The point $x_0 = 2$ should therefore be a good starting point for computing either of the small zeros by the Newton iteration. Carry out the calculation and explain what happens.
11. Carry out five iterations of the complex Newton's method applied to the complex-valued function $f(z) = 1 + z^2 + e^z$, using the starting value $z_0 = -1 + 4i$.
12. Find the first four zeros of the function $f(z) = 1 + z^2 + e^z$ ordered according to increasing complex absolute values. How do you know these are the *first* four zeros and that you have not missed some?
13. Carry out five iterations of Newton's method (for two nonlinear functions in two variables) on the following system:

$$\begin{cases} f_1(x, y) = 1 + x^2 - y^2 + e^x \cos y \\ f_2(x, y) = 2xy + e^x \sin y \end{cases}$$

Use starting values $x_0 = -1$ and $y_0 = 4$. Is this problem related to Computer Problem 11 above, and do they have similar numerical behavior? Explain.

14. Using Newton's method, find the roots of the nonlinear systems.
 - a. $\begin{cases} 4y^2 + 4y + 52x = 19 \\ 169x^2 + 3y^2 + 111x - 10y = 10 \end{cases}$
 - b. $\begin{cases} x + e^{-1x} + y^3 = 0 \\ x^2 + 2xy - y^2 + \tan(x) = 0 \end{cases}$

3.3 Secant Method

Recall that the Newton iteration is defined by the equation

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (1)$$

One of the drawbacks of Newton's method is that it involves the derivative of the function whose zero is sought. To overcome this disadvantage, a number of methods have been proposed. For example, Steffensen's iteration (Problem 3.2.4, p. 90).

$$x_{n+1} = x_n - \frac{[f(x_n)]^2}{f(x_n + f(x_n)) - f(x_n)}$$

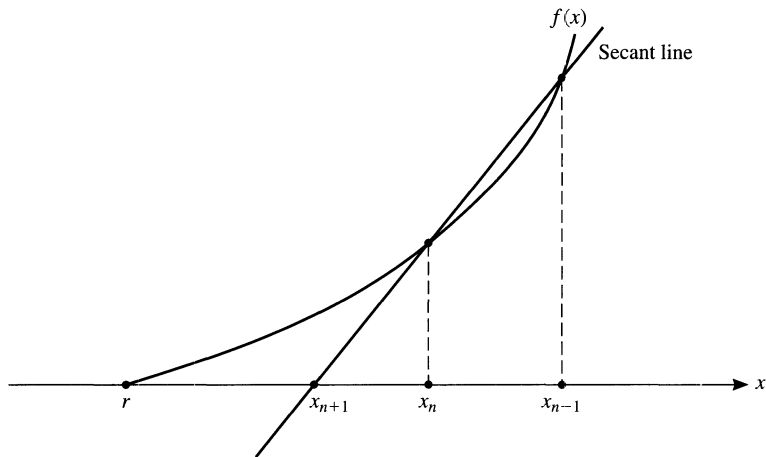


FIGURE 3.6
Geometric interpretation
of secant method

gives one approach to this problem. Another is to replace $f'(x)$ in Equation (1) by a **difference quotient**, such as

$$f'(x_n) \approx \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}} \tag{2}$$

The approximation given in Equation (2) comes directly from the definition of f' as a limit; namely,

$$f'(x) = \lim_{u \rightarrow x} \frac{f(x) - f(u)}{x - u}$$

When this replacement is made, the resulting algorithm is called the **secant method**. Its formula is

$$x_{n+1} = x_n - f(x_n) \left[\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \right] \quad (n \geq 1) \tag{3}$$

Since the calculation of x_{n+1} requires x_n and x_{n-1} , two initial points must be prescribed at the beginning. However, each new x_{n+1} requires only one new evaluation of f . (Steffensen’s algorithm requires two evaluations of f for each new x_{n+1} .)

The graphical interpretation of the secant method is similar to that of Newton’s method. The tangent line to the curve is replaced by a secant line. (See Figure 3.6.)

EXAMPLE 1 Use the secant method to find a zero of the function

$$f(x) = x^3 - \sinh x + 4x^2 + 6x + 9$$

Solution A rough plot suggests that there is a zero between 7 and 8. We take these points as x_0 and x_1 in the algorithm. When this code was run on a machine similar to the

Marc-32, the following results were obtained:

n	x_n	$f(x_n)$
0	8.00000	-0.665×10^3
1	7.00000	0.417×10^2
2	7.05895	0.208×10^2
3	7.11764	-0.183×10^1
4	7.11289	0.710×10^{-1}
5	7.11306	0.244×10^{-3}
6	7.11306	0.191×10^{-4}



Secant Algorithm

An algorithm for the secant method can be written as follows. We modify the standard secant method slightly to obtain nonincreasing function values.

```
input  $a, b, M, \delta, \varepsilon$ 
 $fa \leftarrow f(a); fb \leftarrow f(b)$ 
output 0,  $a, fa$ 
output 1,  $b, fb$ 
for  $k = 2$  to  $M$  do
    if  $|fa| > |fb|$  then
         $a \leftrightarrow b; fa \leftrightarrow fb$ 
    end if
     $s \leftarrow (b - a)/(fb - fa)$ 
     $b \leftarrow a$ 
     $fb \leftarrow fa$ 
     $a \leftarrow a - fa * s$ 
     $fa \leftarrow f(a)$ 

    output  $k, a, fa$ 
    if  $|fa| < \varepsilon$  or  $|b - a| < \delta$  then stop
end do
```

Notice that in the pseudocode the endpoints of $[a, b]$ may be interchanged to keep $|f(a)| \leq |f(b)|$. Thus, the pair $\{x_n, x_{n-1}\}$ has $|f(x_n)| \leq |f(x_{n-1})|$, and the next pair $\{x_{n+1}, x_n\}$ has $|f(x_{n+1})| \leq |f(x_n)|$. This ensures that the absolute value of the function is nonincreasing beginning in step 2.

Error Analysis

The errors in the secant method will now be analyzed. The pseudocode includes a possible switch of the two most recent root guesses. However, we consider only the simple case of no exchanging of endpoints in the following analysis.

From the definition of the secant method (3), we have, with $e_n = x_n - r$,

$$\begin{aligned} e_{n+1} = x_{n+1} - r &= [f(x_n)x_{n-1} - f(x_{n-1})x_n]/[f(x_n) - f(x_{n-1})] - r \\ &= [f(x_n)e_{n-1} - f(x_{n-1})e_n]/[f(x_n) - f(x_{n-1})] \end{aligned}$$

Factoring out $e_n e_{n-1}$ and inserting $(x_n - x_{n-1})/(x_n - x_{n-1})$, we obtain

$$e_{n+1} = \left[\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \right] \left[\frac{f(x_n)/e_n - f(x_{n-1})/e_{n-1}}{x_n - x_{n-1}} \right] e_n e_{n-1} \quad (4)$$

By Taylor's Theorem,

$$f(x_n) = f(r + e_n) = f(r) + e_n f'(r) + \frac{1}{2} e_n^2 f''(r) + \mathcal{O}(e_n^3)$$

Since $f(r) = 0$, this gives us

$$f(x_n)/e_n = f'(r) + \frac{1}{2} e_n f''(r) + \mathcal{O}(e_n^2)$$

Changing the index to $n - 1$ yields

$$f(x_{n-1})/e_{n-1} = f'(r) + \frac{1}{2} e_{n-1} f''(r) + \mathcal{O}(e_{n-1}^2)$$

By subtraction of these equations, we get

$$f(x_n)/e_n - f(x_{n-1})/e_{n-1} = \frac{1}{2} (e_n - e_{n-1}) f''(r) + \mathcal{O}(e_{n-1}^2)$$

Since $x_n - x_{n-1} = e_n - e_{n-1}$, we arrive at the equation

$$\frac{f(x_n)/e_n - f(x_{n-1})/e_{n-1}}{x_n - x_{n-1}} \approx \frac{1}{2} f''(r)$$

The first bracketed expression in Equation (4) can be written as

$$\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \approx \frac{1}{f'(r)}$$

Hence, we have shown that

$$e_{n+1} \approx \frac{1}{2} \frac{f''(r)}{f'(r)} e_n e_{n-1} = C e_n e_{n-1} \quad (5)$$

This equation is similar to the one encountered in the analysis of Newton's method—Equation (3) in Section 3.2 (p. 84). To discover the **order of convergence** of the secant method, we postulate the following asymptotic relationship:

$$|e_{n+1}| \sim A |e_n|^\alpha \quad (6)$$

where A is a positive constant. This means that the ratio $|e_{n+1}|/(A|e_n|^\alpha)$ tends to 1 as $n \rightarrow \infty$ and implies α -order convergence. Hence,

$$|e_n| \sim A |e_{n-1}|^\alpha \quad \text{and} \quad |e_{n-1}| \sim (A^{-1} |e_n|)^{1/\alpha} \quad (7)$$

In Equation (5), we substitute the asymptotic values of $|e_{n+1}|$ and $|e_{n-1}|$ from relations (6) and (7). The result is

$$A|e_n|^\alpha \sim |C| |e_n| A^{-1/\alpha} |e_n|^{1/\alpha}$$

This can be written as

$$A^{1+1/\alpha} |C|^{-1} \sim |e_n|^{1-\alpha+1/\alpha} \tag{8}$$

Since the left side of this relation is a nonzero constant while $e_n \rightarrow 0$, we conclude that $1 - \alpha + 1/\alpha = 0$, or $\alpha = (1 + \sqrt{5})/2 \approx 1.62$, taking the positive root. Hence, the secant method's rate of convergence is **superlinear** (that is, better than linear). Now we can determine A since the right side of relation (8) is 1. Thus, using the equation $1 + 1/\alpha = \alpha$, we have

$$A = |C|^{1/(1+1/\alpha)} = |C|^{1/\alpha} = |C|^{\alpha-1} = |C|^{0.62} = \left| \frac{f''(r)}{2f'(r)} \right|^{0.62}$$

With A as just given, we finally have for the secant method

$$|e_{n+1}| \approx A |e_n|^{(1+\sqrt{5})/2}$$

Since $(1 + \sqrt{5})/2 \approx 1.62 < 2$, the rapidity of convergence of the secant method is not as good as Newton's method but is better than the bisection method. However, each step of the secant method requires only *one* new function evaluation, whereas each step of the Newton algorithm requires *two* function evaluations: $f(x)$ and $f'(x)$. Since function evaluations constitute the principal computational burden in these algorithms, a *pair* of steps in the secant method is comparable to one step in the Newton method. For two steps of the secant method, we have

$$|e_{n+2}| \sim A |e_{n+1}|^\alpha \sim A^{1+\alpha} |e_n|^{\alpha^2} = A^{1+\alpha} |e_n|^{(3+\sqrt{5})/2}$$

This is considerably *better* than the quadratic convergence of the Newton method since $(3 + \sqrt{5})/2 \approx 2.62$. Of course, two steps of the secant method would require more work per iteration.

The three methods we have discussed (bisection, Newton, and secant) illustrate a common phenomenon in numerical analysis and particularly in scientific computing: the trade-off between speed and reliability. Speed is directly related to computing cost. For some computationally intensive problems (such as the numerical solution of partial differential equations), speed is the overriding consideration. For general-purpose software that will be used by many different customers, reliability and robustness are overriding concerns. An algorithm or subprogram is **robust** if it is able to cope with a wide variety of different numerical situations without the intervention of the user. Herculean efforts have been expended over many years to produce general-purpose mathematical libraries that possess both of these attributes. Two fine examples of such libraries are the IMSL [1995] and NAG [1995] collections.

In all scientific computing tasks, established software packages are recommended over programs written by oneself, except for special applications. For the root-finding problem, the best software is rather complicated because it must incorporate several procedures to secure both global convergence and locally rapid convergence. These goals conflict somewhat with each other, as mentioned previously.

Unfortunately, there are too many methods, and variations of them, to include them all in this book. Examples of algorithms of interest that we do not discuss are ones developed by Brent [1973], Dekker [1969], and Le [1985]. They combine the good features of the bisection and secant methods and assume nothing about the function whose root is sought, other than $f(a)f(b) \leq 0$. The idea behind Brent's method is to combine the bisection method with the secant method and include an inverse quadratic interpolation to get a more robust procedure. Le's algorithm combines the bisection method with second- and third-order methods that use derivative estimates from the objective function values. The interested reader is encouraged to refer to the papers cited above for a complete description of these algorithms. Since the resulting codes are rather long and complicated, we recommend obtaining the software for them via the Internet. Our general advice, to use established and proven software whenever possible, holds true here. Usually such software is freely available and has been carefully written and tested. For example, using a browser to the World Wide Web, we can connect to the address <http://gams.nist.gov>, which is a "Guide to Available Mathematical Software." A problem decision tree is found there, and we go down it from *F. Solution of Nonlinear Equations to F1. Single Equation to F1b. Nonpolynomial*. There we find that on Netlib a version of Brent's algorithm in the C-language (for finding a minimum or a zero of a univariate function within a given range) is available. (See also Appendix A, "An Overview of Mathematical Software.")

PROBLEMS 3.3

- 1. Establish Equation (4).
- 2. In the secant method, prove that if $x_n \rightarrow q$ as $n \rightarrow \infty$, and if $f'(q) \neq 0$, then q is a zero of f .
- 3. Using Taylor expansions for $f(x + h)$ and $f(x + k)$, derive the following approximation to $f'(x)$:

$$f'(x) \approx \frac{k^2 f(x + h) - h^2 f(x + k) + (h^2 - k^2) f(x)}{(k - h)kh}$$
- 4. If the secant method is applied to the function $f(x) = x^2 - 2$, with $x_0 = 0$ and $x_1 = 1$, what is x_2 ?
- 5. What is x_2 if $x_0 = 1$, $x_1 = 2$, $f(x_0) = 2$, and $f(x_1) = 1.5$ in an application of the secant method?
- 6. The relation of asymptotic equality between two sequences is written $x_n \sim y_n$ and signifies that $\lim_{n \rightarrow \infty} (x_n / y_n) = 1$. Prove that if $x_n \sim y_n$, $u_n \sim v_n$, and $c \neq 0$, then
 - a. $cx_n \sim cy_n$
 - b. $x_n^c \sim y_n^c$
 - c. $x_n u_n \sim y_n v_n$

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- d. if $y_n \sim u_n$, then $x_n \sim v_n$
 - e. $y_n \sim x_n$
7. Prove that the formula for the secant method can be written in the form

$$x_{n+1} = \frac{f(x_n)x_{n-1} - x_nf(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

Explain why this formula is inferior to Equation (3) in practice.

8. (A polynomial of very high degree) An **annuity** is a fund of money to which payments (not necessarily equal in value) are made at regular intervals of time. In one form of annuity, the fund is invested at a constant rate of interest, r , per interval. The interest is compounded at the end of each interval. Let the payments be in amounts $a_1, a_2, \dots$. Let V_i denote the accumulated value of the annuity, computed just after payment a_i has been made. Then $V_1 = a_1$ and

$$V_i = V_{i-1}(1 + r) + a_i \qquad (i = 2, 3, \dots)$$

The factor $(1 + r)$ accounts for the interest rV_{i-1} earned on V_{i-1} during one time interval. Prove that $V_n = \sum_{i=1}^n a_i x^{n-i}$, where $x = 1 + r$. (Computer Problem 3.3.6, p. 100, is related.)

9. Two men follow two different savings strategies over a period of 44 years. The first saves \$1000 per year for six years and leaves the account to earn interest for the remaining 38 years. The second invests nothing in the first six years, but thereafter saves \$1000 per year. At the end of 44 years, the values of the two accounts are the same. Assume the earnings in both accounts are compounded annually at the same rate of interest. What is the rate and what is the value of each account?
10. How does the interchange in the secant method pseudocode affect the error analysis? Write out the details for a modified error analysis.

COMPUTER
PROBLEMS 3.3

1. Write a subprogram to carry out the secant method on a function f , assuming that two starting points are given. Test the routine on these functions.
- a. $\sin(x/2) - 1$
 - b. $e^x - \tan x$
 - c. $x^3 - 12x^2 + 3x + 1$
2. Program and test a refinement of the secant method that uses the approximate value of $f'(x)$ given in Problem 3.3.3 (p. 98). That is, use this approximation for $f'(x)$ in the formula for the Newton method. Three starting points are now needed. The first two can be arbitrary, and the third can be obtained by the secant method.
3. Write subprograms for carrying out the bisection method, Newton's method, and the secant method. They should apply to an arbitrary function F . In each case, the calling sequence should include a parameter M for the maximum number of steps the user will allow. The user should be able to specify also the accuracy desired (ε and δ , as in the pseudocode in this section). These codes should be in single precision.

- a. Test your subprograms on the function

$$f(x) = \tan^{-1} x - \frac{2x}{1+x^2}$$

Try to obtain the positive zero with full machine precision.

- b. Use the zero found in part a as the starting point for Newton's method on the function

$$g(x) = \tan^{-1} x$$

- c. Combine two of the programs that you have written to create a hybrid method that has good global and local characteristics.
4. Select a routine from the program library available on your computer for solving an equation $f(x) = 0$, without using derivatives. Test the code on these functions on the intervals given.
- a. $x^{20} - 1$ on $[0, 10]$
 - b. $\tan x - 30x$ on $[1, 1.57]$
 - c. $x^2 - (1-x)^{10}$ on $[0, 1]$
 - d. $x^3 + 10^{-4}$ on $[-0.75, 0.5]$
 - e. $x^{19} + 10^{-4}$ on $[-0.75, 0.5]$
 - f. x^5 on $[-1, 10]$
 - g. x^9 on $[-1, 10]$
 - h. xe^{-x^2} on $[-1, 4]$
- (See Nerinckx and Haegemans [1976].)
5. Program and test the secant algorithm using the same example as in the text. Then repeat the computation using 3 and 10 as the initial points. Explain what happened.
6. (Refer to Problem 3.3.8, p. 99.) A succession of 60 monthly payments is made into an annuity. In the first to fifth years, the payments are, respectively, \$200, \$275, \$312, \$380, and \$400. Just after the last payment has been made, the accumulated value of the annuity is \$24,738. What was the monthly rate of interest? Use the secant method to find the zero of the polynomial that arises.

3.4 Fixed Points and Functional Iteration

Newton's method and Steffensen's method are examples of procedures whereby a sequence of points is computed by a formula of the form

$$x_{n+1} = F(x_n) \quad (n \geq 0) \tag{1}$$

The algorithm defined by such an equation is called **functional iteration**. In Newton's method, the function F is given by

$$F(x) = x - \frac{f(x)}{f'(x)}$$

whereas in Steffensen’s method, we have

$$F(x) = x - \frac{[f(x)]^2}{f(x + f(x)) - f(x)}$$

Many other instances of the general procedure of functional iteration could be cited, and we will study the general theory briefly.

Formula (1) can be used to generate sequences that do not converge—for example, the sequence 1, 3, 9, 27, . . . , which arises if $x_0 = 1$ and $F(x) = 3x$. However, our interest lies mainly in those cases where $\lim_{n \rightarrow \infty} x_n$ exists. Then suppose that

$$\lim_{n \rightarrow \infty} x_n = s$$

What is the relation between s and F ? If F is continuous, then

$$F(s) = F\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} F(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = s$$

Thus, $F(s) = s$, and we call s a **fixed point** of the function F . We could think of a fixed point as a value that the function “locks onto” in the iterative process.

Often a mathematical problem can be reduced to the problem of finding a fixed point of a function. Very interesting applications occur in differential equations, optimization theory, and other areas. Usually the function F whose fixed points are sought will be a mapping from one vector space into another. We intend to analyze the simplest case when F maps some closed set $C \subseteq \mathbb{R}$ into itself. The theorem to be proved concerns **contractive mappings**. A mapping (or function) F is said to be **contractive** if there exists a number λ less than 1 such that

$$|F(x) - F(y)| \leq \lambda |x - y| \tag{2}$$

for all points x and y in the domain of F . As illustrated in Figure 3.7, the distance between x and y is mapped into a smaller distance between $F(x)$ and $F(y)$ by the contractive function F .

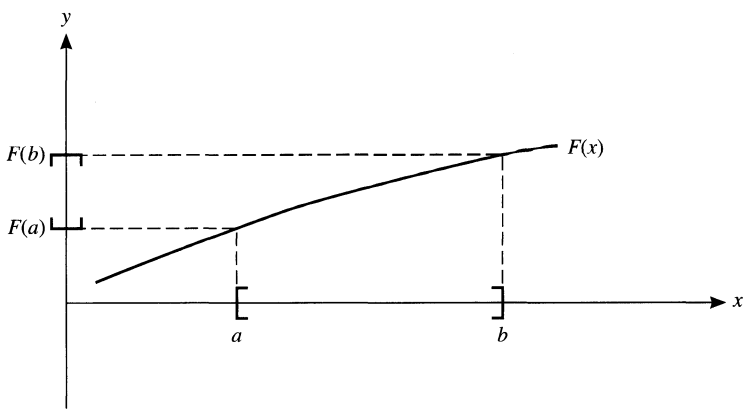


FIGURE 3.7
Graphical example of
contractive mapping

THEOREM 1 Contractive Mapping Theorem

Let C be a closed subset of the real line. If F is a contractive mapping of C into C , then F has a unique fixed point. Moreover, this fixed point is the limit of every sequence obtained from Equation (1) with a starting point $x_0 \in C$.

Proof We use the contractive property (2) together with Equation (1) to write

$$|x_n - x_{n-1}| = |F(x_{n-1}) - F(x_{n-2})| \leq \lambda |x_{n-1} - x_{n-2}| \quad (3)$$

This argument can be repeated to get

$$|x_n - x_{n-1}| \leq \lambda |x_{n-1} - x_{n-2}| \leq \lambda^2 |x_{n-2} - x_{n-3}| \leq \cdots \leq \lambda^{n-1} |x_1 - x_0|$$

Since x_n can be written in the form

$$x_n = x_0 + (x_1 - x_0) + (x_2 - x_1) + \cdots + (x_n - x_{n-1})$$

we see that the sequence $[x_n]$ converges if and only if the series

$$\sum_{n=1}^{\infty} (x_n - x_{n-1})$$

converges. To prove that this series converges, it suffices to prove that the series

$$\sum_{n=1}^{\infty} |x_n - x_{n-1}|$$

converges. This is easy because we can use the comparison test and the previous work:

$$\sum_{n=1}^{\infty} |x_n - x_{n-1}| \leq \sum_{n=1}^{\infty} \lambda^{n-1} |x_1 - x_0| = \frac{1}{1-\lambda} |x_1 - x_0|$$

Since the sequence converges, let $s = \lim_{n \rightarrow \infty} x_n$. Then $F(s) = s$, as noted previously. (Observe that the contractive property implies continuity of F .) As for the unicity of the fixed point, if x and y are fixed points, then

$$|x - y| = |F(x) - F(y)| \leq \lambda |x - y|$$

Since $\lambda < 1$, $|x - y| = 0$. Finally, note that the point s that is obtained belongs to C since s is the limit of a sequence lying in C . ■

Theorem 1 can be proved for a contractive map of any complete metric space into itself.

EXAMPLE 1 Show that the sequence $[x_n]$ in the Contractive Mapping Theorem satisfies the **Cauchy criterion** for convergence.

Solution Recall the Cauchy criterion for the sequence $[x_n]$: Given any ε , there exists an integer N such that $|x_n - x_m| < \varepsilon$ whenever $n, m \geq N$. If $n \geq m \geq N$, then by the triangle inequality and the equation following (3),

$$\begin{aligned} |x_n - x_m| &\leq |x_n - x_{n-1}| + |x_{n-1} - x_{n-2}| + \cdots + |x_{m+1} - x_m| \\ &\leq \lambda^{n-1}|x_1 - x_0| + \lambda^{n-2}|x_1 - x_0| + \cdots + \lambda^m|x_1 - x_0| \\ &= \lambda^m|x_1 - x_0|(1 + \lambda + \lambda^2 + \cdots + \lambda^{n-1-m}) \\ &\leq \lambda^N|x_1 - x_0|(1 + \lambda + \lambda^2 + \cdots) \\ &= \lambda^N|x_1 - x_0|(1 - \lambda)^{-1} \end{aligned}$$

For any $\varepsilon > 0$, there exists an N such that $|x_n - x_m| < \varepsilon$ whenever $n, m \geq N$. We just increase N until $\lambda^N|x_1 - x_0|(1 - \lambda)^{-1} < \varepsilon$. ■

EXAMPLE 2 Prove that the sequence $[x_n]$ defined recursively as follows is convergent.

$$\begin{cases} x_0 = -15 \\ x_{n+1} = 3 - \frac{1}{2}|x_n| \end{cases} \quad (n \geq 0)$$

Solution The function $F(x) = 3 - \frac{1}{2}|x|$ is a contraction because

$$|F(x) - F(y)| = \left| 3 - \frac{1}{2}|x| - 3 + \frac{1}{2}|y| - |x| \right| = \frac{1}{2}||y| - |x|| \leq \frac{1}{2}|y - x|$$

by the triangle inequality. By Theorem 1, the sequence described must converge to the unique fixed point of F , which is readily seen to be 2. In the theorem, we can take C to be $\mathbb{R}$. ■

EXAMPLE 3 Use the Contractive Mapping Theorem to compute a fixed point of the function

$$F(x) = 4 + \frac{1}{3} \sin 2x$$

Solution By the Mean-Value Theorem, we have

$$|F(x) - F(y)| = \frac{1}{3}|\sin 2x - \sin 2y| = \frac{2}{3}|\cos 2\zeta| |x - y| \leq \frac{2}{3}|x - y|$$

for some ζ between x and y . This shows that F is contractive, with $\lambda = 2/3$. By Theorem 1, F has a fixed point. A computer program to compute this fixed point can be based on the following algorithm, which carries out 20 iterations starting with an initial value of 4:

```
input  $x \leftarrow 4$ ;  $M \leftarrow 20$ 
for  $k = 1$  to  $M$  do
     $x \leftarrow 4 + \frac{1}{3} \sin 2x$ 
    output  $k, x$ 
end do
```

The program produces 20 lines of output; several are shown here:

k	x
1	4.32978 61
2	4.23089 51
3	4.27363 38
$\vdots$	$\vdots$
14	4.26148 30
15	4.26148 40
16	4.26148 36
$\vdots$	$\vdots$
20	4.26148 37

The fixed point is given to seven decimal digits of accuracy in the last line. ■

Error Analysis

Now let us analyze the errors in the process of functional iteration. We suppose that F has a fixed point, s , and that a sequence $[x_n]$ has been defined by the formula $x_{n+1} = F(x_n)$. Let

$$e_n = x_n - s$$

If F' exists and is continuous, then by the Mean-Value Theorem,

$$x_{n+1} - s = F(x_n) - F(s) = F'(\zeta_n)(x_n - s)$$

or

$$e_{n+1} = F'(\zeta_n)e_n$$

where ζ_n is a point between x_n and s . The condition $|F'(x)| < 1$ for all x ensures that the errors decrease in magnitude. If e_n is small, then ζ_n is near s , and $F'(\zeta_n) \approx F'(s)$. One would expect rapid convergence if $F'(s)$ is small. An ideal situation would be $F'(s) = 0$. In that case, an additional term in the Taylor series will be useful. To handle several cases at once, let us suppose that q is an integer such that

$$F^{(k)}(s) = 0 \quad \text{for} \quad 1 \leq k < q \quad \text{but} \quad F^{(q)}(s) \neq 0$$

By the Taylor series for $F(x_n)$ expanded about s , we have

$$\begin{aligned} e_{n+1} &= x_{n+1} - s = F(x_n) - F(s) \\ &= F(s + e_n) - F(s) \\ &= \left[F(s) + e_n F'(s) + \frac{1}{2} e_n^2 F''(s) + \cdots \right] - F(s) \\ &= e_n F'(s) + \frac{1}{2} e_n^2 F''(s) + \cdots + \frac{1}{(q-1)!} e_n^{q-1} F^{(q-1)}(s) + \frac{1}{q!} e_n^q F^{(q)}(\zeta_n) \end{aligned}$$

and therefore

$$e_{n+1} = \frac{1}{q!} e_n^q F^{(q)}(\zeta_n) \tag{4}$$

If we know that $\lim_{n \rightarrow \infty} x_n = s$, then Equation (4) implies that

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^q} = \frac{1}{q!} F^{(q)}(s) \tag{5}$$

Recall from Section 1.2 (p. 17) that for any sequence $[x_n]$ converging to a point s (whether it arises from a functional iteration or not), we define the **order of convergence** to be the largest real number q such that the limit

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1} - s|}{|x_n - s|^q}$$

exists and is nonzero. Such a q need not exist, and when it does, it need not be an integer.

If, for example, $F'(s) = 0$ and $F''(s) \neq 0$, then $q = 2$ and we have

$$e_{n+1} = \frac{1}{2} e_n^2 F''(\zeta_n)$$

This is reminiscent of Newton's method, Equation (3) in Section 3.2 (p. 84). In fact, Newton's method uses

$$F(x) = x - \frac{f(x)}{f'(x)}$$

and for it we have

$$F'(x) = 1 - \frac{f'(x)f'(x) - f(x)f''(x)}{[f'(x)]^2} = \frac{f(x)f''(x)}{[f'(x)]^2}$$

Since a fixed point of F is a zero of f , we have $F'(s) = 0$. Moreover,

$$F''(x) = \frac{[f'(x)]^2[f(x)f'''(x) + f''(x)f'(x)] - [f(x)f''(x)][2f'(x)f''(x)]}{[f'(x)]^4}$$

and therefore we usually have

$$F''(s) = \frac{f''(s)}{f'(s)} \neq 0$$

If F is represented near a fixed point s by a Taylor series

$$\sum_{k=0}^{\infty} \frac{1}{k!} F^{(k)}(s)(x-s)^k$$

and if $x_{n+1} = F(x_n)$, then the order of convergence is the first integer q such that $F^{(q)}(s) \neq 0$. Problem 3.4.16 (p. 107) asks for a proof of this assertion.

PROBLEMS 3.4

- 1. If F is contractive from $[a, b]$ to $[a, b]$ and $x_{n+1} = F(x_n)$, with $x_0 \in [a, b]$, then $|x_n - s| \leq C\lambda^n$ for an appropriate C . Prove this and give an upper bound for C . Here s is the fixed point of F .
- 2. Prove that if $F: [a, b] \rightarrow \mathbb{R}$, if F' is continuous, and if $|F'(x)| < 1$ on $[a, b]$, then F is a contraction. Does F necessarily have a fixed point?
- 3. Prove that if F is a continuous map of $[a, b]$ into $[a, b]$, then F must have a fixed point. Then determine whether this assertion is true for functions from $\mathbb{R}$ to $\mathbb{R}$.
- 4. Show that these functions are contractive on the indicated intervals. Determine the best values of λ in Equation (2).
 - a. $(1 + x^2)^{-1}$ on an arbitrary interval
 - b. $\frac{1}{2}x$ on $1 \leq x \leq 5$
 - c. $\tan^{-1} x$ on an arbitrary closed interval excluding 0
 - d. $|x|^{\frac{3}{2}}$ on $|x| \leq \frac{1}{3}$
- 5. Kepler's equation in astronomy reads $x = y - \varepsilon \sin y$, with $0 < \varepsilon < 1$. Show that for each $x \in [0, \pi]$, there is a y satisfying the equation. Interpret this as a fixed-point problem.
- 6. Consider an iteration function of the form $F(x) = x + f(x)g(x)$, where $f(r) = 0$ and $f'(r) \neq 0$. Find the precise conditions on the function g so that the method of functional iteration will **converge cubically** to r if started near r .
- 7. If you enter a number into a handheld calculator and then repeatedly press the cosine button, what number will eventually appear? Provide a proof.
- 8. Prove that the sequence generated by the iteration $x_{n+1} = F(x_n)$ will converge if $|F'(x)| \leq \lambda < 1$ on the interval $[x_0 - \rho, x_0 + \rho]$, where $\rho = |F(x_0) - x_0|/(1 - \lambda)$.
- 9. What special properties must a function f have if Newton's method applied to f converges cubically to a zero of f ?
- 10. If we attempt to find a fixed point of F by using Newton's method on the equation $F(x) - x = 0$, what iteration formula results?
- 11. If f' is continuous and positive on $[a, b]$, and if $f(a)f(b) < 0$, then f has *exactly* one zero in (a, b) . Prove this, and show that with a suitable parameter λ , the zero can be obtained by applying the method of functional iteration, with $F(x) = x + \lambda f(x)$.
- 12. Let p be a positive number. What is the value of the following expression?

$$x = \sqrt{p + \sqrt{p + \sqrt{p + \cdots}}}$$

Note that this can be interpreted as meaning $x = \lim_{n \rightarrow \infty} x_n$, where $x_1 = \sqrt{p}$, $x_2 = \sqrt{p + \sqrt{p}}$, and so forth. *Hint:* Observe that $x_{n+1} = \sqrt{p + x_n}$.

- 13. (Continuation) Let $p > 1$. What is the value of the following continued fraction?

$$x = \frac{1}{p + \frac{1}{p + \frac{1}{p + \cdots}}}$$

Use the ideas of the preceding problem to solve this one. Prove that the sequence of values converges by using the Contractive Mapping Theorem.

14. (Continuation) Solve for the roots of the quadratic equation $x^2 + px + q = 0$ by developing an iterative method.
15. Let F be a contractive mapping of an interval $[a, b]$ into itself, and let s be the fixed point of F . If $a \leq x \leq b$ and $|F(x) - x| < \varepsilon$, does it follow that $|x - s| < \varepsilon$? Prove that $|x - s| < \varepsilon(1 - \lambda)^{-1}$, where λ is the constant in Equation (2).
16. Prove the statement at the end of this section in the text concerning the order of convergence in the method of functional iteration.
17. Most iterative processes are not as simple as the one expressed by $x_{n+1} = F(x_n)$, with $F: \mathbb{R} \rightarrow \mathbb{R}$. For example, we might have a map $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Show that the bisection method and the secant method are of this type. In each case, define F explicitly.
18. Prove that if F' is continuous and if $|F'(x)| < 1$ on the interval $[a, b]$, then F is contractive on $[a, b]$. Show that this is not necessarily true for an open interval.
19. If the method of functional iteration is applied to the function $F(x) = 2 + (x - 2)^4$, starting at $x = 2.5$, what order of convergence results? Find the range of starting values for which this functional iteration converges. Note that 2 is a fixed point.
20. Show that the following functions are contractive on the given domains, yet they have no fixed point on these domains. Why does this not contradict the Contractive Mapping Theorem?
 - a. $F(x) = 3 - x^2$ on $[-\frac{1}{4}, \frac{1}{4}]$
 - b. $F(x) = -x/2$ on $[-2, -1] \cup [1, 2]$
21. Prove that if f is continuous on $[a, b]$ and satisfies $a \leq f(a)$ and $f(b) \leq b$, then f has a fixed point in the interval $[a, b]$. Note that we do not assume $a \leq f(x) \leq b$ for all x in $[a, b]$.
22. What is the weakest condition that can be put on the interval $[c, d]$ so that each continuous map of $[a, b]$ into $[c, d]$ shall have a fixed point?
23. Find the order of convergence of these sequences.
 - a. $x_n = (1/n)^{\frac{1}{2}}$
 - b. $x_n = \sqrt[n]{n}$
 - c. $x_n = (1 + 1/n)^{\frac{1}{2}}$
 - d. $x_{n+1} = \tan^{-1} x_n$
24. To find a zero of the function f , we can look for a fixed point of the function $F(x) = x - f(x)/f'(x)$. To find a fixed point of F , we can solve $F(x) - x = 0$ by Newton's method. When this is done, what is the formula for generating the sequence x_n ?
25. Prove that the function F defined by $F(x) = 4x(1 - x)$ maps the interval $[0, 1]$ into itself and is not a contraction. Prove that it has a fixed point. Why does this not contradict the Contractive Mapping Theorem?
26. If the method of functional iteration is used on $f(x) = \frac{1}{2}(1 + x^2)^{-1}$ starting at $x_0 = 7$, will the resulting sequence converge? If so, what is the limit? Establish your answers rigorously.
27. Prove or disprove: If $F: \mathbb{R} \rightarrow [a, b]$, and if F is contractive on $[a, b]$, then F has a unique fixed point, which can be obtained by the method of functional iteration, starting at any real value.
28. Give examples of functions that *do not* have fixed points but *do* have these characteristics:
 - a. $f: [0, 1] \rightarrow [0, 1]$

- b. $f: (0, 1) \rightarrow (0, 1)$ and is continuous
 - c. $f: A \rightarrow A$ and is continuous, with $A = [0, 1] \cup [2, 3]$
 - d. $f: \mathbb{R} \rightarrow \mathbb{R}$ and is continuous
29. Prove that the function $f(x) = 2 + x - \tan^{-1} x$ has the property $|f'(x)| < 1$. Prove that f does not have a fixed point. Explain why this does not contradict the Contractive Mapping Theorem.
30. This problem concerns the function $F(x) = 10 - 2x$. Prove that F has a fixed point. Let x_0 be arbitrary, and define $x_{n+1} = F(x_n)$ for $n \geq 0$. Find a nonrecursive formula for x_n . Prove that the method of functional iteration does not produce a convergent sequence unless x_0 is given a particular value. What is this special value for x_0 ? Why does this not contradict the Contractive Mapping Theorem?
31. Let F be continuously differentiable in an open interval, and suppose that F has a fixed point s in this open interval. Prove that if $|F'(s)| < 1$, then the sequence defined by functional iteration will converge to s if started sufficiently close to s . *Hint:* Select λ so that $|F'(r)| < \lambda < 1$, and consider an interval centered at r in which $|F'(x)| < \lambda$.
32. Let $\frac{1}{2} \leq q \leq 1$, and define $F(x) = 2x - qx^2$. On what interval can it be guaranteed that the method of iteration using F will converge to a fixed point? (This problem is related to Problem 3.2.5, p. 90.)
33. Write down two different fixed-point procedures for finding a zero of the function $f(x) = 2x^2 + 6e^{-x} - 4$.
34. On which of these intervals $[\frac{1}{2}, \infty]$, $[\frac{1}{8}, 1]$, $[\frac{1}{4}, 2]$, $[0, 1]$, $[\frac{1}{5}, \frac{3}{2}]$ is the function $f(x) = \sqrt{x}$ contractive?
35. A function F is called an **iterated contraction** if

$$|F(F(x)) - F(x)| \leq \lambda |F(x) - x| \quad (\lambda < 1)$$

Show that every contraction is an iterated contraction. Show that an iterated contraction need not be a contraction nor continuous.

36. If the method of functional iteration is used on $F(x) = x^2 + x - 2$ and produces a convergent sequence of positive numbers, what is the limit of that sequence and what was the starting point?
37. Consider a function of the form $F(x) = x - f(x)f'(x)$, where $f(r) = 0$ and $f'(r) \neq 0$. Find the precise conditions on the function f so that the method of functional iteration will converge at least **cubically** to r if started near r .
38. Analyze Steffensen's method (Problem 3.2.4, p. 90) as an example of functional iteration. Determine its order of convergence.
39. A student incorrectly recalls Newton's method and writes $x_{n+1} = f(x_n)/f'(x_n)$. Will this method find a zero of f ? What is the order of convergence?
40. Show that the following method has third-order convergence for computing $\sqrt{R}$:

$$x_{n+1} = \frac{x_n(x_n^2 + 3R)}{3x_n^2 + R}$$

41. Consider an iterative method of the form $x_{n+1} = x_n - f(x_n)/g(x_n)$. Assume that it converges to a point ξ that is a simple zero of the function f but not a zero of the function g . Establish the relationship between f and g so that the order of convergence of the method is 3 or greater.

3.5 Computing Roots of Polynomials

Methods developed in the preceding sections—in particular, Newton’s method—can certainly be applied to polynomials. But in finding the roots of a polynomial, we should exploit the special structure of such a function, if possible. Moreover, the polynomial problem is frequently complicated by our wish to compute the complex roots (even if the polynomial has real coefficients) or all the roots of a given polynomial. Thus, the topic of finding the roots of a polynomial deserves special treatment, which indeed it has received for almost 400 years!

We begin with some important theoretical results, most of which are probably familiar to the reader. We write a polynomial in the form

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_2 z^2 + a_1 z + a_0 \tag{1}$$

in which the coefficients a_k and the variable z may be complex numbers. If $a_n \neq 0$, then p has **degree** n . We are interested in finding the *roots* of p . Are there any? The following theorem (first proved by Gauss in 1799) answers this question. The proof given here from 1938 is by G. H. Hardy [1960, Appendix I] and Fefferman [1967].

THEOREM 1 Fundamental Theorem of Algebra

Every nonconstant polynomial has at least one root in the complex field.

Proof Let p be a nonconstant polynomial. We want to show that $p(z_0) = 0$ for some $z_0 \in \mathbb{C}$. Since p is not constant, $|p(z)| \rightarrow \infty$ when $|z| \rightarrow \infty$. Let D be a disk centered at 0 outside of which $|p(z)| \geq |p(0)|$. Let z_0 be a point where $\inf_{z \in D} |p(z)|$ is attained. Since $0 \in D$, $|p(z_0)| \leq |p(0)|$. Thus, $|p(z_0)| \leq |p(z)|$ for all $z \in \mathbb{C}$. Put $q(z) = p(z + z_0)$. We want to prove that $q(0) = 0$, so that $p(z_0) = 0$. Write $q(z) = c_0 + c_j z^j + \cdots + c_n z^n = c_0 + c_j z^j + z^{j+1} r(z)$, in which $c_j \neq 0$ and r is a polynomial (possibly 0). Now we want to prove that $c_0 = 0$. Suppose $c_0 \neq 0$. Let w be any complex number such that $c_j w^j = -c_0$. Define $N = \sup_{0 < \varepsilon < 1} |r(\varepsilon w)|$. Select ε in $(0, 1)$ so small that $\varepsilon |w|^{j+1} N < |c_0|$. Then we obtain a contradiction as follows:

$$\begin{aligned} |q(\varepsilon w)| &\leq |c_0 + c_j \varepsilon^j w^j| + \varepsilon^{j+1} |w|^{j+1} |r(\varepsilon w)| \\ &= |c_0 - c_0 \varepsilon^j| + \varepsilon^j \varepsilon |w|^{j+1} N \\ &< |c_0| (1 - \varepsilon^j) + \varepsilon^j |c_0| = |c_0| = |q(0)| \\ &= |p(z_0)| \leq |p(z_0 + \varepsilon w)| = |q(\varepsilon w)| \end{aligned} \quad \blacksquare$$

The history of this theorem and of attempts to prove it can be read in the book by Kline [1972, pp. 597–606]. The usual *modern* proof is as a corollary to Liouville’s Theorem. See, for example, Bak and Newman [1982], Henrici [1974], Ahlfors [1966], or other texts on complex analysis.

The Fundamental Theorem of Algebra does *not* assert the existence of real roots, and the simplest examples, such as $z^2 + 1$, show that a polynomial need not have real roots, even if its coefficients are all real.

If the polynomial p , having degree n at least 1, is divided by a linear factor $z - c$, the result is a quotient q and a remainder r . The latter is a complex number, and the former is a polynomial of degree $n - 1$. We can represent the process by the equation

$$p(z) = (z - c)q(z) + r$$

From this we see (by letting $z = c$) that $p(c) = r$. This fact is known as the **Remainder Theorem**. If c is a root of p , then $r = 0$, and we have

$$p(z) = (z - c)q(z)$$

Thus, $z - c$ is a factor of $p(z)$. This implication is known as the **Factor Theorem**.

Let us write $p(z) = (z - r_1)q_1(z)$, where r_1 is any root of p . By the Fundamental Theorem of Algebra, q_1 has a root, say r_2 (unless q_1 is of degree 0). Hence, $q_1(z)$ has a factor $z - r_2$, and we can write $p(z) = (z - r_1)(z - r_2)q_2(z)$. Continuing in this manner, we eventually arrive at a stopping point because the degrees of the successive q_k decrease by 1 at each step. Hence, q_n will be a constant, and the last equation is

$$p(z) = (z - r_1)(z - r_2) \cdots (z - r_n)q_n$$

This proves that a polynomial of degree n has a factorization into a product of n linear factors, each corresponding to a root of p . It is clear that p can have no other roots because if z is any complex number different from $r_1, r_2, \dots, r_n$, then the product $\prod_{k=1}^n (z - r_k)$ cannot be 0. Since some of the roots r_k may be equal to each other, we see that a polynomial of degree n can have at most n roots. The **multiplicity** of a root is the number of times that the factor corresponding to it appears in the factorization of p . If we combine the preceding remarks with the Fundamental Theorem of Algebra, we obtain the following theorem:

THEOREM 2 Theorem on Complex Roots of Polynomials

A polynomial of degree n has exactly n roots in the complex plane, it being agreed that each root shall be counted a number of times equal to its multiplicity.

It is often desirable to know roughly where the roots of a polynomial are situated in the complex plane. Many results, such as Descartes' rule of signs, are available in the literature for this problem of **localizing the roots**. See, for example, Section 5.5 of Young and Gregory [1972], Section 5.5 of Stoer and Bulirsch [1980], the treatise of Marden [1966], or Volume 1 of the three-volume treatise of Henrici [1974]. The following theorem, adapted from the textbook of Conte and de Boor [1980], provides an easily computed upper bound for the moduli of the roots.

THEOREM 3 Localization Theorem

All roots of the polynomial in Equation (1) lie in the open disk whose center is at the origin of the complex plane and whose radius is

$$\rho = 1 + |a_n|^{-1} \max_{0 \leq k < n} |a_k|$$

Proof Put $c = \max_{0 \leq k < n} |a_k|$ so that $c|a_n|^{-1} = \rho - 1$. If $c = 0$, our result is trivially true. Hence, assume $c > 0$. Then $\rho > 1$. If $|z| \geq \rho$, then (because $\rho > 1$)

$$\begin{aligned} |p(z)| &\geq |a_n z^n| - |a_{n-1} z^{n-1} + \cdots + a_0| \\ &\geq |a_n z^n| - c \sum_{k=0}^{n-1} |z|^k \\ &> |a_n z^n| - c|z|^n(|z| - 1)^{-1} \\ &= |a_n z^n| \{1 - c|a_n|^{-1}(|z| - 1)^{-1}\} \\ &\geq |a_n z^n| \{1 - c|a_n|^{-1}(\rho - 1)^{-1}\} = 0 \end{aligned}$$

EXAMPLE 1 Find a disk centered at the origin that contains all the roots of the polynomial

$$p(z) = z^4 - 4z^3 + 7z^2 - 5z - 2$$

Solution By Theorem 3, one such disk has radius

$$\rho = 1 + |a_4|^{-1} \max_{0 \leq k < 4} |a_k| = 8$$

Another useful idea available for the analysis of polynomials is this: Take the polynomial p of Equation (1), and consider the function $s(z) = z^n p(1/z)$. Then

$$\begin{aligned} s(z) &= z^n \left[a_n \left(\frac{1}{z}\right)^n + a_{n-1} \left(\frac{1}{z}\right)^{n-1} + \cdots + a_0 \right] \\ &= a_n + a_{n-1}z + a_{n-2}z^2 + \cdots + a_0z^n \end{aligned}$$

This shows that s is a polynomial of degree at most n . Its coefficients can be written down at once, since they are the same as the coefficients of p except in the reverse order. Now it is obvious that for a nonzero complex number z_0 , the condition $p(z_0) = 0$ is equivalent to the condition $s(1/z_0) = 0$. Thus, we have the following result.

THEOREM 4 Theorem on Localization of Roots

If all the roots of s are in the disk $\{z : |z| \leq \rho\}$, then all the nonzero roots of p are outside the disk $\{z : |z| < \rho^{-1}\}$.

EXAMPLE 2 Find a disk centered at the origin in which there are no roots of p , where p is the polynomial of Example 1.

Solution The polynomial s referred to in Theorem 4 is

$$s(z) = -2z^4 - 5z^3 + 7z^2 - 4z + 1$$

So by Theorem 3, all roots of s lie in a disk centered at the origin having radius $\rho = 1 + |a_4|^{-1} \max_{0 \leq k < 4} |a_k| = 9/2$. Therefore, by Theorem 4, the roots of p lie outside the disk of radius $2/9$. Hence, all roots of p lie in the ring $2/9 < |z| < 8$ in the complex plane.

Horner’s Algorithm

For the efficient computation of values of a polynomial, **Horner’s algorithm** is needed. This algorithm is also known as **nested multiplication** and as **synthetic division**. We shall find that it is useful for other purposes as well. If a polynomial p and a complex number z_0 are given, Horner’s algorithm will produce the number $p(z_0)$ and the polynomial

$$q(z) = \frac{p(z) - p(z_0)}{z - z_0} \tag{2}$$

The polynomial q is of degree 1 less than the degree of p . From this equation we have

$$p(z) = (z - z_0)q(z) + p(z_0) \tag{3}$$

Let the unknown polynomial q be represented by

$$q(z) = b_0 + b_1z + \cdots + b_{n-1}z^{n-1}$$

When this form for $q(z)$ and the analogous form for $p(z)$ are substituted into Equation (3), the coefficients of like powers of z on the two sides of the equation can be set equal to each other. These equations arise from doing so:

$$\begin{aligned} b_{n-1} &= a_n \\ b_{n-2} &= a_{n-1} + z_0b_{n-1} \\ &\vdots \\ b_0 &= a_1 + z_0b_1 \\ p(z_0) &= a_0 + z_0b_0 \end{aligned}$$

In compact form, Horner’s algorithm can be written like this:

```
input  $n, (a_i : 0 \leq i \leq n), z_0$ 
 $b_{n-1} \leftarrow a_n$ 
for  $k = n - 1$  to 0 step  $-1$  do
     $b_{k-1} \leftarrow a_k + z_0b_k$ 
end do
output  $(b_i : -1 \leq i \leq n - 1)$ 
```

Notice that $b_{-1} = p(z_0)$ in this pseudocode. If the calculation in Horner’s algorithm is to be carried out with pencil and paper, the following arrangement is often used:

	a_n	a_{n-1}	a_{n-2}	$\cdots$	a_0
z_0		z_0b_{n-1}	z_0b_{n-2}	$\cdots$	z_0b_0
	b_{n-1}	b_{n-2}	b_{n-3}	$\cdots$	b_{-1}

The boxed number satisfies $p(z_0) = b_{-1}$.

EXAMPLE 3 Use Horner’s algorithm to evaluate $p(3)$, where p is the polynomial

$$p(z) = z^4 - 4z^3 + 7z^2 - 5z - 2$$

Solution We arrange the calculation as suggested above.

	1	-4	7	-5	-2
3		3	-3	12	21
	1	-1	4	7	19

Thus $p(3) = 19$, and we can write

$$p(z) = (z - 3)(z^3 - z^2 + 4z + 7) + 19$$

Horner’s algorithm is used also for **deflation**. This is the process of removing a linear factor from a polynomial. If z_0 is a root of the polynomial p , then $z - z_0$ is a factor of p , and the converse is true. The remaining roots of p are the $n - 1$ roots of $p(z)/(z - z_0)$.

EXAMPLE 4 Deflate the polynomial p of Example 3 using the fact that 2 is one of its roots.

Solution We use the same arrangement of computations as explained previously:

	1	-4	7	-5	-2
2		2	-4	6	2
	1	-2	3	1	0

Thus, we have

$$z^4 - 4z^3 + 7z^2 - 5z - 2 = (z - 2)(z^3 - 2z^2 + 3z + 1)$$

A third application of Horner’s algorithm is in finding the Taylor expansion of a polynomial about any point. Let $p(z)$ be as in Equation (1), and suppose that we desire the coefficients c_k in the equation

$$\begin{aligned} p(z) &= a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0 \\ &= c_n (z - z_0)^n + c_{n-1} (z - z_0)^{n-1} + \cdots + c_0 \end{aligned}$$

Of course, Taylor’s Theorem asserts that $c_k = p^{(k)}(z_0)/k!$, but we seek a more efficient algorithm. Notice that $p(z_0) = c_0$, so that this coefficient is obtained by applying Horner’s algorithm to the polynomial p with the point z_0 . The algorithm also yields the polynomial

$$q(z) = \frac{p(z) - p(z_0)}{z - z_0} = c_n (z - z_0)^{n-1} + c_{n-1} (z - z_0)^{n-2} + \cdots + c_1$$

This shows that the second coefficient, c_1 , can be obtained by applying Horner’s algorithm to the polynomial q with point z_0 because $c_1 = q(z_0)$. (Notice that the

first application of Horner’s algorithm does not yield q in the form shown but rather as a sum of powers of z .) This process is repeated until all coefficients c_k are found.

EXAMPLE 5 Find the Taylor expansion of the polynomial in Example 3, about the point $z_0 = 3$.

Solution The work can be arranged as follows:

	1	−4	7	−5	−2	
3			3	−3	12	21
	1	−1	4	7		19
3			3	6	30	
	1	2	10			37
3			3	15		
	1	5				25
3			3			
	1					8

The boxed numbers are used in the polynomial

$$p(z) = (z - 3)^4 + 8(z - 3)^3 + 25(z - 3)^2 + 37(z - 3) + 19$$
 ■

We call the algorithm just described the **complete Horner’s algorithm**. The pseudocode for executing it is arranged so that the coefficients c_k *overwrite* the input coefficients a_k .

```
input n, (ai : 0 ≤ i ≤ n), z0
for k = 0 to n − 1 do
    for j = n − 1 to k step − 1 do
        aj ← aj + z0aj+1
    end do
end do
output (ai : 0 ≤ i ≤ n)
```

We have everything necessary for showing how Newton’s iteration can be carried out on a polynomial. Recall that this iteration is defined by the equation

$$z_{k+1} = z_k - \frac{f(z_k)}{f'(z_k)}$$

If this is applied to a polynomial p , an efficient method incorporating Horner’s algorithm can be used to compute $p(z)$ and $p'(z)$. We saw previously that if only two steps inside the complete Horner’s algorithm are used, we obtain $c_0 = p(z_0)$ and $c_1 = p'(z_0)$. These two steps can be combined in the pseudocode. Also, we desist from overwriting the input coefficients because they will be required in subsequent

steps of the iteration. Here is a pseudocode that produces $\alpha = p(z_0)$ and $\beta = p'(z_0)$, assuming that p is given as in Equation (1) and that z_0 is given:

```
input  $n, (a_i : 0 \leq i \leq n), z_0$ 
 $\alpha \leftarrow a_n$ 
 $\beta \leftarrow 0$ 
for  $k = n - 1$  to  $0$  step  $-1$  do
     $\beta \leftarrow \alpha + z_0\beta$ 
     $\alpha \leftarrow a_k + z_0\alpha$ 
end do
output  $\alpha, \beta$ 
```

If we refer to this pseudocode as `horner` ($n, (a_i : 0 \leq i \leq n), z_0, \alpha, \beta$), then a pseudocode for taking M steps in Newton’s method on the given polynomial, starting at z_0 , would look like this:

```
input  $n, (a_i : 0 \leq i \leq n), z_0, M, \varepsilon$ 
for  $j = 1$  to  $M$  do
    call horner ( $n, (a_i : 0 \leq i \leq n), z_0, \alpha, \beta$ )
     $z_1 \leftarrow z_0 - \alpha/\beta$ 
    output  $\alpha, \beta, z_1$ 
    if  $|z_1 - z_0| < \varepsilon$  stop
     $z_0 \leftarrow z_1$ 
end do
```

EXAMPLE 6 Carry out Newton’s iteration on the polynomial used in Example 3, starting at $z_0 = 0$.

Solution First, we use $z_0 = 0$, and we compute the values $p(0) = -2$ and $p'(0) = -5$ by the algorithms previously explained. The new value of z is

$$z_1 = z_0 - \frac{p(z_0)}{p'(z_0)} = 0 - \frac{-2}{-5} = -0.4$$

Executing the algorithm on a computer with the precision of the `MarC-32` produces the following results:

k	$p(z_k)$	$p'(z_k)$	z_k
1	-2.00000	-5.00000	-0.40000
2	1.40160	-12.77600	-0.29029
3	1.46322	-10.17322	-0.27591
4	0.00226	-9.86030	-0.27568
5	0.00000	-9.85537	-0.27568

Observe that z_k converges rapidly to the root -0.27568 . ■

A formal proof that Horner’s algorithm is correct follows.

THEOREM 5 Theorem on Horner's Method

Let $p(x) = a_n x^n + \cdots + a_1 x + a_0$. Define pairs (α_j, β_j) for $j = n, n-1, \dots, 0$ by the algorithm

$$\begin{cases} (\alpha_n, \beta_n) = (a_n, 0) \\ (\alpha_j, \beta_j) = (a_j + x\alpha_{j+1}, \alpha_{j+1} + x\beta_{j+1}) \quad (n-1 \geq j \geq 0) \end{cases}$$

Then $\alpha_0 = p(x)$ and $\beta_0 = p'(x)$.

Proof If $n = 0$, then $p(x) = a_0$ and

$$(\alpha_0, \beta_0) = (a_0, 0)$$

So obviously, $\alpha_0 = p(x)$ and $\beta_0 = p'(x)$. Thus, the theorem is true for $n = 0$. If $n = 1$, then $p(x) = a_1 x + a_0$ and

$$(\alpha_0, \beta_0) = (a_0 + x\alpha_1, \alpha_1 + x\beta_1) = (a_0 + xa_1, a_1) = (p(x), p'(x))$$

Thus, the theorem is true for $n = 1$. Suppose the theorem is true for all indices n less than m . Let

$$p(x) = c_m x^m + \cdots + c_1 x + c_0 = c_0 + x(c_m x^{m-1} + \cdots + c_2 x + c_1) = c_0 + xq(x)$$

Consequently, we have

$$p'(x) = xq'(x) + q(x)$$

Set $n = m - 1$, $a_n = c_m$, $a_{n-1} = c_{m-1}$, $\dots$, $a_2 = c_3$, $a_1 = c_2$, and $a_0 = c_1$. Since the theorem is true for q , we apply the algorithm to $q(x) = a_n x^n + \cdots + a_1 x + a_0$. We get $\alpha_0 = q(x)$ and $\beta_0 = q'(x)$ by the induction hypothesis. If the algorithm is applied to p , we get the same set of pairs $(\alpha_n, \beta_n), \dots, (\alpha_1, \beta_1), (\alpha_0, \beta_0)$ (because $(c_m, \dots, c_2, c_1) = (a_n, \dots, a_1, a_0)$), but there is one more pair to be computed at the end for p ; namely,

$$\begin{aligned} (\alpha_{-1}, \beta_{-1}) &= (a_{-1} + x\alpha_0, \alpha_0 + x\beta_0) \\ &= (c_0 + xq(x), q(x) + xq'(x)) \\ &= (p(x), p'(x)) \end{aligned}$$

■

Note that we could define our sequence by

$$\begin{cases} (\alpha_{n-1}, \beta_{n-1}) = (a_{n-1} + xa_n, a_n) \\ (\alpha_j, \beta_j) = (a_j + x\alpha_{j+1}, \alpha_{j+1} + x\beta_{j+1}) \quad (n-2 \geq j \geq 0) \end{cases}$$

or by

$$\begin{cases} (\alpha_n, \beta_n) = (a_{n-1} + xa_n, a_n) \\ (\alpha_j, \beta_j) = (a_{j-1} + x\alpha_{j+1}, \alpha_{j+1} + x\beta_{j+1}) \quad (n-1 \geq j \geq 1) \end{cases}$$

These are *inferior* because they do not work when $n = 0$.

THEOREM 6 Theorem on Successive Newton Iterates

Let x_k and x_{k+1} be two successive iterates when Newton's method is applied to a polynomial p of degree n . Then there is a root of p within distance $n|x_k - x_{k+1}|$ of x_k in the complex plane.

Proof Let $r_1, r_2, \dots, r_n$ be the roots of p . Then $p(z) = c \prod_{j=1}^n (z - r_j)$. The correction term in the Newton iteration is $-p(z)/p'(z)$. The derivative of p is

$$p'(z) = c \sum_{k=1}^n \prod_{\substack{i=1 \\ i \neq k}}^n (z - r_i) = \sum_{k=1}^n p(z)/(z - r_k) = p(z) \sum_{k=1}^n (z - r_k)^{-1}$$

We want to show that for any z (playing the role of x_k) there is an index j for which $|z - r_j| \leq n|p(z)/p'(z)|$. If no index j satisfies the desired inequality, then for all j , $|z - r_j| > n|p(z)/p'(z)|$. From this it would follow that

$$|z - r_j|^{-1} < \frac{1}{n} |p'(z)/p(z)| = \frac{1}{n} \left| \sum_{k=1}^n (z - r_k)^{-1} \right| \leq \frac{1}{n} \sum_{k=1}^n |z - r_k|^{-1}$$

But this is not possible because the average of n numbers cannot be greater than each of them. ■

This theorem is due to Bodewig [1946].

Bairstow's Method

If a polynomial has only real coefficients, its roots may nevertheless be complex. In this case, it is possible to compute the complex roots two at a time using only real arithmetic. The procedure for doing this is called **Bairstow's method**; it will be considered next.

For any complex number $z = x + iy$, the **conjugate** number is $\bar{z} = x - iy$. A basic result that will be exploited follows.

THEOREM 7 Theorem on Real Quadratic Factor

If p is a polynomial whose coefficients are all real, and if w is a nonreal root of p , then $\bar{w}$ is also a root, and $(z - w)(z - \bar{w})$ is a real quadratic factor of p .

Proof Let $p(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$, with all a_k being real. Since w is a root of p , we have

$$0 = a_n w^n + a_{n-1} w^{n-1} + \dots + a_1 w + a_0$$

Take the conjugate of both sides, using repeatedly the rules $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ and $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$. Since the a_k are real, the result is

$$0 = a_n \bar{w}^n + a_{n-1} \bar{w}^{n-1} + \dots + a_1 \bar{w} + a_0$$

Thus, $\bar{w}$ is a root of p . Since w is not real, w and $\bar{w}$ are distinct roots of p . Consequently, p contains the quadratic factor

$$(z - w)(z - \bar{w}) = z^2 - (w + \bar{w})z + w\bar{w}$$

This factor is real because $w + \bar{w}$ and $w\bar{w}$ are real. ■

The nonreal roots of the real polynomial p occur in conjugate pairs, and these pairs induce quadratic real factors of p . Hence, it is reasonable to search for quadratic factors; this can be done with the aid of Newton's iteration.

THEOREM 8 Theorem on Quotient and Remainder

If the polynomial $p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0$ is divided by the quadratic polynomial $z^2 - uz - v$, then the quotient and remainder

$$\begin{aligned} q(z) &= b_n z^{n-2} + b_{n-1} z^{n-3} + \cdots + b_3 z + b_2 \\ r(z) &= b_1(z - u) + b_0 \end{aligned}$$

can be computed recursively by setting $b_{n+1} = b_{n+2} = 0$ and then using

$$b_k = a_k + ub_{k+1} + vb_{k+2} \quad (n \geq k \geq 0)$$

Proof The relationship between p , q , and r is expressed by

$$p(z) = q(z)(z^2 - uz - v) + r(z)$$

In detail, this equation reads:

$$\sum_{k=0}^n a_k z^k = \left(\sum_{k=2}^n b_k z^{k-2} \right) (z^2 - uz - v) + b_1(z - u) + b_0$$

If we equate the coefficients of z^k on the two sides of this equation, the result is

$$\begin{aligned} a_k &= b_k - ub_{k+1} - vb_{k+2} \quad (0 \leq k \leq n-2) \\ a_{n-1} &= b_{n-1} - ub_n \\ a_n &= b_n \end{aligned}$$

The first of these three equations will include the last two if we define b_{n+1} and b_{n+2} to be 0. ■

Let us specialize the analysis to the case in which all the coefficients a_k in p are real. We shall seek a real quadratic factor of the type mentioned in the theorem; thus, u and v will be real. In the division process above, b_0 and b_1 are functions of u and v , and we write $b_0 = b_0(u, v)$ and $b_1 = b_1(u, v)$. In order that q be a factor of p , the remainder r should vanish; this leads to the two equations

$$b_0(u, v) = 0$$

$$b_1(u, v) = 0$$

In Bairstow's method, this pair of simultaneous nonlinear equations is solved by Newton's method. We require the partial derivatives

$$c_k = \frac{\partial b_k}{\partial u} \quad d_k = \frac{\partial b_{k-1}}{\partial v} \quad (0 \leq k \leq n)$$

These are obtained by differentiating the recurrence relation already established for b_k in Theorem 8. The result is the following pair of additional recurrences:

$$\begin{aligned} c_k &= b_{k+1} + uc_{k+1} + vc_{k+2} & (c_{n+1} = c_n = 0) \\ d_k &= b_{k+1} + ud_{k+1} + vd_{k+2} & (d_{n+1} = d_n = 0) \end{aligned} \quad (4)$$

Since these recurrence relations obviously generate the same two sequences, we need only the first. An outline of the process is as follows: Starting values are assigned to u and v . We seek corrections, denoted by δu and δv , so that the equations

$$b_0(u + \delta u, v + \delta v) = b_1(u + \delta u, v + \delta v) = 0$$

are true. As in Section 3.2 (p. 88), we linearize these equations by writing

$$\begin{aligned} b_0(u, v) + \frac{\partial b_0}{\partial u} \delta u + \frac{\partial b_0}{\partial v} \delta v &= 0 \\ b_1(u, v) + \frac{\partial b_1}{\partial u} \delta u + \frac{\partial b_1}{\partial v} \delta v &= 0 \end{aligned}$$

In view of the preceding remarks, this system becomes

$$\begin{bmatrix} c_0 & c_1 \\ c_1 & c_2 \end{bmatrix} \begin{bmatrix} \delta u \\ \delta v \end{bmatrix} = - \begin{bmatrix} b_0(u, v) \\ b_1(u, v) \end{bmatrix}$$

The solution of this system follows:

$$\begin{aligned} \delta u &= (c_1 b_1 - c_2 b_0) / J \\ \delta v &= (c_1 b_0 - c_0 b_1) / J \\ J &= c_0 c_2 - c_1^2 \end{aligned}$$

Notice that J is the Jacobian determinant for the pair of nonlinear functions $b_0(u, v)$ and $b_1(u, v)$.

A pseudocode to execute M steps of Bairstow's method, starting at a prescribed point (u, v) , is given next. It follows the preceding description closely.

```

input  $n, (a_i : 0 \leq i \leq n), u, v, M$ 
 $b_n \leftarrow a_n$ 
 $c_n \leftarrow 0$ 
 $c_{n-1} \leftarrow a_n$ 
for  $j = 1$  to  $M$  do
     $b_{n-1} \leftarrow a_{n-1} + ub_n$ 
    for  $k = n - 2$  to  $0$  step  $-1$  do
         $b_k \leftarrow a_k + ub_{k+1} + vb_{k+2}$ 

```

```

         $c_k \leftarrow b_{k+1} + uc_{k+1} + vc_{k+2}$ 
    end do
     $J \leftarrow c_0c_2 - c_1^2$ 
     $u \leftarrow u + (c_1b_1 - c_2b_0)/J$ 
     $v \leftarrow v + (c_1b_0 - c_0b_1)/J$ 
    output  $j, u, v, b_0, b_1$ 
end do

```

EXAMPLE 7 Find a real quadratic factor of the polynomial in the previous examples, using Bairstow's method and starting with the point $(u, v) = (3, -4)$.

Solution A computer program based on the pseudocode above was written in double precision, and at the seventh iteration, we found

$$u = 2.27568\ 22036\ 510$$

$$v = -3.62736\ 50847\ 118$$

$$b_0 = -0.2 \times 10^{-14}$$

$$b_1 = 0.0$$

Since b_0 and b_1 are practically 0, we can accept u and v as the approximate values of the coefficients in the quadratic factor $z^2 - uz - v$. By combining the results of this and previous examples, we can exhibit the factorization of the polynomial p . (We display only three-digit precision in the equation.)

$$z^4 - 4z^3 + 7z^2 - 5z - 2 = (z - 2)(z + 0.276)(z^2 - 2.28z + 3.63)$$

The values of the coefficients have been given previously with higher accuracy. Two complex roots of p can be computed from the quadratic factor (using its more accurate coefficients); they are

$$1.13784\ 11018\ 255 \pm (1.52731\ 22508\ 866)i$$

■

To complete the analysis of the Bairstow method, we need to establish (under reasonable hypotheses) that the Jacobian J is not 0 at the solution point.

THEOREM 9 Theorem on the Jacobian in Bairstow's Method

Let (u_0, v_0) be a point such that the roots of $z^2 - u_0z - v_0$ are simple roots of p . Then the Jacobian in Bairstow's method is not 0 at (u_0, v_0) .

Proof At every step in the process, we have

$$p(z) = (z^2 - uz - v)q(z) + b_1(z - u) + b_0$$

Partial differentiation with respect to u and v gives us these two equations:

$$0 = -zq(z) + (z^2 - uz - v)\frac{\partial q}{\partial u} - b_1 + \frac{\partial b_1}{\partial u}(z - u) + \frac{\partial b_0}{\partial u}$$

$$0 = -q(z) + (z^2 - uz - v)\frac{\partial q}{\partial v} + \frac{\partial b_1}{\partial v}(z - u) + \frac{\partial b_0}{\partial v}$$

Assuming the hypotheses, we know that $z^2 - u_0z - v_0$ has two roots, say z_1 and z_2 . It follows that $p(z_1) = 0$, $p(z_2) = 0$, $b_1 = 0$, and $b_0 = 0$. In the preceding equations, let $u = u_0$, $v = v_0$, and $z = z_1$ or $z = z_2$. Four equations result from this:

$$0 = -z_jq(z_j) + c_1(z_j - u_0) + c_0 \quad (j = 1, 2)$$

$$0 = -q(z_j) + c_2(z_j - u_0) + c_1 \quad (j = 1, 2)$$

Here we use the notation and analysis that immediately follow the proof of Theorem 8. In matrix form, these equations look like this:

$$\begin{bmatrix} c_0 & c_1 \\ c_1 & c_2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ z_1 - u_0 & z_2 - u_0 \end{bmatrix} = \begin{bmatrix} z_1q(z_1) & z_2q(z_2) \\ q(z_1) & q(z_2) \end{bmatrix}$$

To see that the Jacobian matrix is not singular, it suffices to prove that the matrix on the right of the preceding equation is not singular. The determinant of this matrix is $(z_1 - z_2)q(z_1)q(z_2)$. Since z_1 and z_2 are simple roots of p , they cannot be roots of q . Also, $z_1 \neq z_2$, and therefore this determinant is not 0. ■

The discussion of Bairstow's method follows the exposition of Henrici [1964].

Laguerre Iteration

We turn next to Laguerre's method for finding the roots of a polynomial p . This method is used in several modern software packages because it has very favorable convergence properties and is rather robust. It exhibits third-order convergence in the vicinity of each simple root. The algorithm is iterative and proceeds from one approximate root z to a new one by calculating

$$A = -p'(z)/p(z)$$

$$B = A^2 - p''(z)/p(z)$$

$$C = n^{-1} [A \pm \sqrt{(n-1)(nB - A^2)}]$$

$$z_{\text{new}} = z + 1/C$$

Here p is a polynomial of degree n . In the definition of C , the sign is chosen to make $|C|$ as large as possible.

For this algorithm, the following theorem of Kahan [1967] is the analogue of Theorem 6.

THEOREM 10 Theorem on Radius of Convergence in Laguerre's Method

If p is a polynomial of degree n , if z is any complex number, and if C is computed as in Laguerre's algorithm, then p has a root in the complex plane within distance $\sqrt{n}/C$ of z .

Proof As in Theorem 6, on successive Newton iterates, we denote the roots of our polynomial by $r_1, r_2, \dots, r_n$. We can assume that $p(x) = \prod_{j=1}^n (x - r_j)$, with leading coefficient unity, since the formulas for A , B , and C are not affected by any scalar factor attached to p . Differentiating p and putting $u_j = (r_j - z)^{-1}$ yield

$$p'(z) = \sum_{j=1}^n \prod_{\substack{k=1 \\ k \neq j}}^n (z - r_k) = \sum_{j=1}^n p(z)/(z - r_j) = -p(z) \sum_{j=1}^n u_j$$

Hence, we have

$$A = -\frac{p'(z)}{p(z)} = \sum_{j=1}^n u_j$$

Differentiating in this formula and using the definition of B lead to

$$B = \frac{-p(z)p''(z) + [p'(z)]^2}{[p(z)]^2} = \sum_{j=1}^n (r_j - z)^{-2} = \sum_{j=1}^n u_j^2$$

From the definition of C , we have

$$(nC - A)^2 = (n - 1)(nB - A^2) \quad (5)$$

Define $D = (A - C)/(n - 1)$, so that

$$A = (n - 1)D + C \quad (6)$$

From Equation (5), it follows that

$$\begin{aligned} n^2 C^2 - 2nCA + A^2 &= (n - 1)nB - nA^2 + A^2 \\ (n - 1)B &= nC^2 - 2CA + A^2 \end{aligned} \quad (7)$$

In Equation (7), replace A by $(n - 1)D + C$ to get

$$\begin{aligned} (n - 1)B &= nC^2 - 2C[(n - 1)D + C] + [(n - 1)D + C]^2 \\ (n - 1)B &= nC^2 - 2(n - 1)CD - 2C^2 + (n - 1)^2 D^2 + 2(n - 1)CD + C^2 \\ (n - 1)B &= (n - 1)C^2 + (n - 1)^2 D^2 \\ B &= C^2 + (n - 1)D^2 \end{aligned} \quad (8)$$

Starting with (7) and then using (6) and (8) yield

$$\begin{aligned} nB - A^2 &= nC^2 - 2CA + B \\ &= nC^2 - 2C[(n - 1)D + C] + C^2 + (n - 1)D^2 \\ &= (n - 1)C^2 - (n - 1)2CD + (n - 1)D^2 \\ &= (n - 1)(C - D)^2 \end{aligned} \quad (9)$$

Starting with Equation (9) and using formulas for A and B , we obtain

$$\begin{aligned} (n - 1)|C - D|^2 &= |nB - A^2| \\ &= n^{-1}|n^2 B - 2nA^2 + nA^2| \end{aligned}$$

$$\begin{aligned}
 &= n^{-1} \left| \sum_{j=1}^n (n^2 u_j^2 - 2n A u_j + A^2) \right| \\
 &= n^{-1} \left| \sum_{j=1}^n (n u_j - A)^2 \right| \\
 &\leq n^{-1} \sum_{j=1}^n |n u_j - A|^2 \\
 &= n^{-1} \sum_{j=1}^n \{n^2 |u_j|^2 - 2n \operatorname{Re}(\bar{A} u_j) + |A|^2\} \\
 &= n \sum_{j=1}^n |u_j|^2 - 2\bar{A} A + |A|^2 \\
 &\leq n^2 \max_j |u_j|^2 - |A|^2 \\
 &= n^2 \max_j |u_j|^2 - |(n-1)D + C|^2 \\
 &= n^2 \max_j |u_j|^2 - |C - D + nD|^2 \\
 &= n^2 \max_j |u_j|^2 - |C - D|^2 - 2n \operatorname{Re}[\bar{D}(C - D)] - n^2 |D|^2
 \end{aligned}$$

Hence, we have

$$\begin{aligned}
 n|C - D|^2 &\leq n^2 \max_j |u_j|^2 - n^2 |D|^2 - 2n \operatorname{Re}(\bar{D}C) + 2n|D|^2 \\
 |C - D|^2 &\leq n \max_j |u_j|^2 - n|D|^2 - 2 \operatorname{Re}(\bar{D}C) + 2|D|^2 \\
 |C|^2 - 2 \operatorname{Re}(C\bar{D}) + |D|^2 &\leq n \max_j |u_j|^2 - n|D|^2 - 2 \operatorname{Re}(\bar{D}C) + 2|D|^2 \\
 |C|^2 + (n-1)|D|^2 &\leq n \max_j |u_j|^2 = n / \min_j |z - r_j|^2
 \end{aligned}$$

This gives

$$\min_j |z - r_j|^2 \leq n / \{|C|^2 + (n-1)|D|^2\}$$

and

$$\min_j |z - r_j| \leq \sqrt{n} / \sqrt{|C|^2 + (n-1)|D|^2} \quad (10)$$

This inequality is stronger than the one in the statement of the theorem. The proof is from Kahan [1967]. ■

A simple pseudocode to execute Laguerre's method on a given polynomial with a prescribed starting point z_0 is offered next:

```

input  $n, (a_i : 0 \leq i \leq n), z_0, M, \varepsilon$ 
for  $k = 1$  to  $M$  do
     $\alpha \leftarrow a_n$ 
     $\beta \leftarrow 0$ 
     $\gamma \leftarrow 0$ 
    for  $j = n - 1$  to  $0$  step  $-1$  do
         $\gamma \leftarrow z_0\gamma + \beta$ 
         $\beta \leftarrow z_0\beta + \alpha$ 
         $\alpha \leftarrow z_0\alpha + a_j$ 
    end do
     $A \leftarrow -\beta/\alpha$ 
     $B \leftarrow A^2 - 2\gamma/\alpha$ 
     $C \leftarrow [A \pm \sqrt{(n-1)(nB - A^2)}]/n$ 
     $z_1 \leftarrow z_0 + 1/C$ 
    output  $k, z$ 
    if  $|z_1 - z_0| < \varepsilon$  then stop
     $z_0 \leftarrow z_1$ 
end do

```

In this algorithm, α is $p(z_0)$, β is $p'(z_0)$, and γ is $\frac{1}{2}p''(z_0)$.

LEMMA 1 First Lemma on Interval Endpoints

Let $v_1, v_2, \dots, v_n$ be any real numbers. Put $\alpha = \sum_{i=1}^n v_i$ and $\beta = \sum_{i=1}^n v_i^2$. Then the numbers v_i lie in the closed interval whose endpoints are

$$n^{-1}[\alpha \pm \sqrt{(n-1)(n\beta - \alpha^2)}] \quad (11)$$

Proof It suffices to prove that v_1 lies in the interval described. Recall the Cauchy-Schwarz Inequality:

$$\left(\sum_{i=1}^m x_i y_i\right)^2 \leq \left(\sum_{i=1}^m x_i^2\right) \left(\sum_{j=1}^m y_j^2\right)$$

Applying this, we have

$$\begin{aligned}
 \alpha^2 - 2\alpha v_1 + v_1^2 &= (\alpha - v_1)^2 = (v_2 + v_3 + \dots + v_n)^2 \\
 &\leq (1^2 + 1^2 + \dots + 1^2)(v_2^2 + v_3^2 + \dots + v_n^2) \\
 &= (n-1)(v_2^2 + v_3^2 + \dots + v_n^2) \\
 &= (n-1)(\beta - v_1^2) - (n-1)\beta - nv_1^2 + v_1^2
 \end{aligned}$$

Rearranging this inequality gives us

$$nv_1^2 - 2\alpha v_1 + \alpha^2 - (n-1)\beta \leq 0$$

This shows that the quadratic function $q(x) = nx^2 - 2\alpha x + \alpha^2 - (n-1)\beta$ has the property $q(v_1) \leq 0$. For large $|x|$, obviously $q(x) > 0$. Hence, v_1 lies between the two roots of q , and they are the endpoints in Formula (11). ■

LEMMA 2 Second Lemma on Interval Endpoints

Let p be a real polynomial of degree n , whose roots, $r_1, r_2, \dots, r_n$, are real. For any real x different from all the r_j , the numbers $(x - r_j)^{-1}$ lie in the interval whose endpoints are

$$[np(x)]^{-1} \left\{ p'(x) \pm \sqrt{[(n-1)p'(x)]^2 - n(n-1)p(x)p''(x)} \right\} \quad (12)$$

Proof The polynomial p is of the form $p(x) = c \prod_{j=1}^n (x - r_j)$. Since the assertion to be proved is independent of c , we can assume that $c = 1$. As in the proof of Theorem 10, on radius of convergence of Laguerre's Method, we have, with $v_j = (x - r_j)^{-1}$, $\alpha = \sum_{j=1}^n v_j$, and $\beta = \sum_{j=1}^n v_j^2$,

$$p'(x)/p(x) = \sum_{j=1}^n v_j = \alpha \quad (13)$$

$$[(p'(x))^2 - p(x)p''(x)] / (p(x))^2 = \sum_{j=1}^n v_j^2 = \beta \quad (14)$$

By Lemma 1, all the numbers v_j lie in an interval with endpoints given in Formula (11). When the values of α and β in Equations (13) and (14) are substituted into this formula, the result is Formula (12). ■

THEOREM 11 Theorem on Monotonic Convergence of Laguerre Method

Let p be a real polynomial whose roots are all real. The sequence produced by the Laguerre algorithm with an arbitrary starting point converges monotonically to a root of p .

Proof Let the roots of p be $r_1 \leq r_2 \leq \dots \leq r_n$. Suppose that x is not a root. Then by Lemma 2, all the numbers $(x - r_i)^{-1}$ lie in an interval whose endpoints are

$$u(x) = [p'(x) + w(x)] / np(x) \quad \text{and} \quad v(x) = [p'(x) - w(x)] / np(x)$$

where

$$w(x) = \sqrt{[(n-1)p'(x)]^2 - n(n-1)p''(x)p(x)}$$

First, let us consider the case when $r_j < x < r_{j+1}$ for some j . Suppose that $p(y) > 0$ on (r_j, r_{j+1}) ; the other case is similar. Then we have

$$v(x) \leq (x - r_{j+1})^{-1} < 0 < (x - r_j)^{-1} \leq u(x)$$

From this it follows that

$$r_j \leq x - \frac{1}{u(x)} < x < x - \frac{1}{v(x)} \leq r_{j+1}$$

Thus, if we start with x in (r_j, r_{j+1}) and compute $x - 1/u(x)$ and $x - 1/v(x)$, as in the Laguerre iteration, then these two new points will lie in $[r_j, r_{j+1}]$. Of course, if either of these two new points is an endpoint of this interval, then a root of p has been computed. In the contrary case, the two new points are in (r_j, r_{j+1}) , and the process can be repeated. Proceeding formally, we define two sequences $[y_k]$ and $[z_k]$ by putting

$$\begin{cases} y_0 = x \\ y_{k+1} = y_k - 1/u(y_k) \end{cases} \quad (k \geq 1)$$

and

$$\begin{cases} z_0 = x \\ z_{k+1} = z_k - 1/v(z_k) \end{cases} \quad (k \geq 1)$$

Our analysis above shows that

$$r_j \leq y_{k+1} < y_k < \cdots < x < z_1 < \cdots < z_k < z_{k+1} \leq r_{j+1}$$

Since the sequence $[y_k]$ decreases and is bounded below by r_j , it must converge. Say $y_k \downarrow y$. From the iteration formulas, we have

$$1/u(y_k) = y_k - y_{k+1} \rightarrow 0$$

Hence, $u(y_k) \rightarrow \infty$. The formula for u now indicates that $p(y_k) \rightarrow 0$, so $p(y) = 0$ and $y = r_j$. Similarly, we conclude that $z_k \uparrow r_{j+1}$.

In the remaining case, the starting point is not between two roots of p . Say $x > r_n$. Suppose also that $p(y) > 0$ on (r_n, ∞) . Proceeding as before, we have

$$0 < (x - r_n)^{-1} \leq u(x)$$

From this, it follows that $r_n \leq x - 1/u(x) < x$. The sequence $[y_n]$, defined as before, converges downward to r_n by the same reasoning. ■

Discussions of Laguerre's method can be found in papers and books by Bodewig [1946], van der Corput [1946], Durand [1960], Foster [1981], Galeone [1977], Householder [1970], Kahan [1967], Ostrowski [1966], Parlett [1964], Redish [1974], and Wilkinson [1965].

Complex Newton's Method

For a polynomial having complex coefficients, Newton's method should be programmed in complex arithmetic. After one root has been found, the deflation process (also programmed in complex arithmetic) should be used. Newton's method can then be applied to the reduced polynomial. This process can be repeated until all roots have been determined. Further analysis and experience indicate that in general, the procedure is satisfactory provided that two precautions are taken:

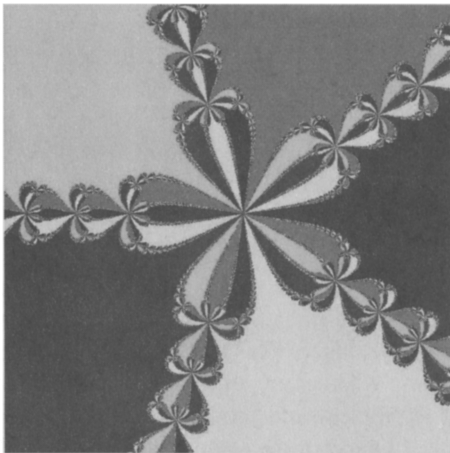


FIGURE 3.8
Basins of attraction for
 $p(x) = z^5 + 1$.

1. The roots should be computed in order of increasing magnitude.
2. Any root obtained by using Newton's method on a reduced polynomial should be immediately refined by applying Newton's method to the original polynomial with the best estimate of the root as the starting value. Only after this has been done should the next step of deflation be carried out.

We recommend that the reader consult Wilkinson [1984] and Peters and Wilkinson [1971] for further guidance on this overall strategy. Many other methods could be discussed for the polynomial root-finding problem. The method of Laguerre is especially attractive because of its favorable global convergence behavior. A robust method suitable for general software has been developed by Jenkins and Traub [1970a]. Other references not already mentioned are Allgower, Glasshoff, and Peitgen [1981], Gautschi [1979, 1984], Henrici [1974], Householder [1970], Ostrowski [1966], Jenkins and Traub [1970b], Marden [1966], Smale [1981], Stoer and Bulirsch [1980], Ralston and Rabinowitz [1978], and Traub [1964].

Here is a brief explanation of how Figure 3.8 was obtained. It involves Newton's method in the complex plane.

Let p be a polynomial of degree at least 2, and let ξ be one of its roots. If Newton's method is started at a point z in the complex plane, it produces a sequence defined by the equations

$$\begin{cases} z_0 = z \\ z_{n+1} = z_n - p(z_n)/p'(z_n) \end{cases} \quad (n \geq 0)$$

If $\lim_{n \rightarrow \infty} z_n = \xi$, we say that z (the starting point) is **attracted** to ξ . The set of all points z that are attracted to ξ is called the **basin of attraction** corresponding to ξ . Each root of p has a basin of attraction, and these are mutually disjoint sets because a sequence that converges to one root of p cannot converge to another root. Some complex numbers belong to no basin of attraction; these are starting points for which the Newton iteration does not converge. These exceptional points constitute the **Julia**

set of p , so named in honor of G. Julia, a French mathematician who published an important memoir on this subject in 1918. If all roots of p are simple, then the basins of attraction are open sets in the complex plane, and the Julia set is the boundary of each basin of attraction.

The basins of attraction for the polynomial $p(z) = z^5 + 1$ are shown in Figure 3.8. The five roots of p are:

$$\omega_k = \cos\left(\frac{2}{5}\pi k\right) + i \sin\left(\frac{2}{5}\pi k\right) \quad k = 0, 1, 2, 3, 4$$

We took a square region in the complex plane and generated a large number of lattice points in that square. For each lattice point, we ran a rough test to determine to which basin of attraction it belonged. This test consisted of monitoring the first 20 Newton iterates and checking whether any of them was within distance 0.25 of a root. If so, then subsequent iteration would produce quadratic convergence to that root. This fact was established by use of the standard convergence theory of Newton’s method in the complex plane. In this way, a list of lattice points belonging to each basin of attraction was generated. The basin of attraction for root ω_0 was assigned one color, and the basins for roots $\omega_1, \omega_2, \omega_3$, and ω_4 were assigned four other colors. These five basins of attraction (actually just the lattice points in them) were displayed on the color screen of a workstation, and then a color plot was produced from it. The five color sets fit together in an incredible way that displays a **fractal** appearance. That is, on magnifying a portion of the plane where two sets meet, we see the same general patterns repeated. This persists on repeated magnification. Furthermore, each boundary point of any one of these five sets is also a boundary point of the other three sets!

Recently, a number of articles and books have been published on fractals and chaos. Additional information can be found, for example, in Barnsley [1988], Curry, Garnett, and Sullivan [1983], Dewdney [1988], Glied [1987], Mandelbrot [1982], Peitgen and Richter [1986], Peitgen, Saupe, and Haeseler [1984], Pickover [1988], and Sander [1987].

PROBLEMS 3.5

- 1. Use Horner’s algorithm to find $p(4)$, where
$$p(z) = 3z^5 - 7z^4 - 5z^3 + z^2 - 8z + 2$$
- 2. (Continuation) For the polynomial of preceding problem, find its expansion in a Taylor series about the point $z_0 = 4$.
- 3. (Continuation) For the polynomial of Problem 3.5.1 (above), start Newton’s method at the point $z_0 = 4$. What is z_1 ?
- 4. (Continuation) For the polynomial of Problem 3.5.1 (above), apply Bairstow’s method with the initial point $(u, v) = (3, 1)$. Compute the corrections δu and δv .
- 5. (Continuation) For the polynomial of Problem 3.5.1 (above), find a disk centered at the origin that contains all the roots.
- 6. (Continuation) For the polynomial of Problem 3.5.1 (above), find a disk centered at the origin that contains none of the roots.
- 7. Does Theorem 3 sometimes give the radius of the smallest circle centered at the origin that contains all the roots of a given polynomial?

- 8. Prove that every polynomial having real coefficients can be factored into a product of linear and quadratic factors having real coefficients.
- 9. Verify the recurrence relations and starting values given for c_k and d_k in the discussion of Bairstow's method.
- 10. For the polynomial $p(z) = 9z^4 - 7z^3 + z^2 - 2z + 5$, compute $p(6)$, $p'(6)$, and the next point in the Newton iteration starting at $z = 6$.
- 11. Using the definition of multiplicity adopted in the text, prove that if z is a root of multiplicity m of a polynomial p , then $p(z) = p'(z) = \cdots = p^{(m-1)}(z) = 0$ and $p^{(m)}(z) \neq 0$.
- 12. (Continuation) Prove the converse of the assertion in the preceding problem.
- 13. Does Bairstow's method yield a quadratically convergent sequence (u_k, v_k) ?
- 14. Write an algorithm that deflates $p(z)$ when a root z_0 is known, but computes the coefficients of the reduced polynomial in ascending order—that is, constant term first.
- 15. Derive Equation (4) in the discussion of Bairstow's method.
- 16. Concerning the polynomial $p(x) = a_0 + a_1x + \cdots + a_nx^n$, prove the following result. For a given x , we set $(\alpha_n, \beta_n, \gamma_n) = (a_n, 0, 0)$ and define inductively

$$(\alpha_j, \beta_j, \gamma_j) = (a_j + x\alpha_{j+1}, \alpha_{j+1} + x\beta_{j+1}, \beta_{j+1} + x\gamma_{j+1})$$

for $j = n - 1, n - 2, \dots, 0$. Then $p(x) = \alpha_0$, $p'(x) = \beta_0$, and $p''(x) = 2\gamma_0$.

- 17. In the analysis of Laguerre's method, prove that

$$C^2 + (n - 1)D^2 = \sum_{j=1}^n u_j^2$$

- 18. In the description of Laguerre's method, the quantities A and B are functions of z and depend on the given polynomial p . Let r be a root of p . Show that the corresponding functions A and B for $p(z)/(z - r)$ are, respectively,

$$A + (z - r)^{-1} \quad \text{and} \quad B - (z - r)^2$$

- 19. Prove that if p is a polynomial of degree n having real coefficients, then

$$(n - 1) [p'(x)]^2 \geq np(x)p''(x)$$

Assume that the roots are real.

COMPUTER
PROBLEMS 3.5

- 1. Write a program that takes as input the coefficients of a polynomial p and a specific point z_0 and produces as output the values $p(z_0)$, $p'(z_0)$, and $p''(z_0)$. Write the pseudocode with only one loop. Test on the polynomial in Problem 3.5.1 (p. 128), taking $z_0 = 4$.
- 2. Write a complex Newton method for a polynomial having complex coefficients, with a given starting point in the complex plane and a given number of iterations. Test your program on the polynomial of Problem 3.5.1 (p. 128), starting with $z_0 = 3 - 2i$.
- 3. Experiment with Laguerre's method coded in complex arithmetic. Find all four roots of the polynomial used in the examples of this section.
- 4. Using Newton's method and the polynomial $p(z) = z^3 - 1$, find three nearby starting points (within 0.01 of each other) so that the resulting sequences converge to different roots. Using a plotter, show the paths of these sequences of points within a square containing the roots by connecting successive points with line segments.

- 5. Write a computer routine that uses Newton’s method and deflation for finding all roots of a polynomial. Test the routine using Wilkinson’s perfidious polynomial given in Computer Problem 2.3.6 (p. 71).
- 6. Write and test a program to compute all the roots of a complex polynomial. Use Laguerre’s method, and compute the roots in order of ascending magnitude. Use **deflation**, and refine any root obtained from a deflated polynomial by Laguerre’s iteration with the original polynomial. To test your program, compute all roots of the polynomial

$$x^8 - 36x^7 + 546x^6 - 4536x^5 + 22449x^4 - 67284x^3 + 118124x^2 - 109584x + 40320$$

The correct roots are the integers 1, 2, . . . , 8. Next, solve the same equation when the coefficient of x^7 is changed to -37 . Observe how a minor perturbation in the coefficients can cause massive changes in the roots. Thus, the roots are unstable functions of the coefficients.

- 7. Modify the Laguerre algorithm so that the coefficients in the quotient polynomials are saved in the application of the Horner algorithm. A separate calculation of the deflated polynomial will then be unnecessary after a root has been computed. Test your program on
- 8. Write a computer program using the complex Newton’s method to compute the basins of attraction of the roots of $f(z) = z^8 + 1$, as discussed in this section. Display the results on a graphics computer terminal or workstation.

$$z^4 - (10 + 26i)z^3 - (216 - 190i)z^2 + (1140 + 636i)z - (72 - 68i) = 0$$

3.6 Homotopy and Continuation Methods

In this section, we consider the problem of finding the roots of an equation

$$f(x) = 0 \tag{1}$$

Here f is a mapping from one linear space to another, say $f : X \rightarrow Y$. This problem is sufficiently general to include systems of algebraic equations, integral equations, differential equations, and so on. The methods to be considered here involve a somewhat different strategy from that in the preceding sections, and this new approach requires the numerical solution of systems of ordinary differential equations, a topic not taken up in this book until Section 8.6 (p. 565). Also, one of our illustrations of the method involves the linear programming problem, which is also taken up later, in Section 10.3 (p. 695).

Basic Concepts

The basic idea of the continuation method is to embed the given problem in a one-parameter family of problems, using a parameter t that runs over the interval $[0, 1]$. The original problem is made to correspond to $t = 1$, and another problem with known solution is made to correspond to $t = 0$. For example, we can define

$$h(t, x) = tf(x) + (1 - t)g(x) \tag{2}$$

The equation $g(x) = 0$ should have a known solution. The next step is to select points $t_0, t_1, \dots, t_m$ so that

$$0 = t_0 < t_1 < t_2 < \dots < t_m = 1$$

We then attempt to solve each equation $h(t_i, x) = 0$, $0 \leq i \leq m$. Assuming that some iterative method will be used (such as Newton's method), it is prudent to use the solution at the i th step as the starting point in computing a solution at the $(i+1)$ st step.

This whole procedure can be regarded as a cure for the difficulty that besets Newton's method: the need for a good starting point.

The relationship (2), which embeds the original problem (1) in a family of problems, gives an example of a **homotopy** that connects the two functions f and g . In general, a homotopy can be *any* continuous connection between f and g . Formally, a homotopy between two functions $f, g : X \rightarrow Y$ is a continuous map

$$h : [0, 1] \times X \rightarrow Y \quad (3)$$

such that $h(0, x) = g(x)$ and $h(1, x) = f(x)$. If such a map exists, we say that f is **homotopic** to g . This is an equivalence relation among the continuous maps from X to Y , where X and Y can be any two topological spaces.

A simple homotopy that is often used in the continuation method is

$$\begin{aligned} h(t, x) &= tf(x) + (1 - t)[f(x) - f(x_0)] \\ &= f(x) + (t - 1)f(x_0) \end{aligned} \quad (4)$$

Here x_0 can be any point in X , and it is clear that x_0 will be a solution of the problem when $t = 0$.

If the equation $h(t, x) = 0$ has a unique root for each $t \in [0, 1]$, then that root is a function of t , and we can write $x(t)$ as the unique member of X that makes the equation $h(t, x(t)) = 0$ true. The set

$$\{x(t) : 0 \leq t \leq 1\} \quad (5)$$

can be interpreted as an arc or curve in X , parametrized by t . This arc leads from the known point $x(0)$ to the solution of our problem, $x(1)$. The continuation method attempts to determine this curve by computing points on it, $x(t_0), x(t_1), \dots, x(t_m)$.

If the function $t \mapsto x(t)$ is differentiable and if h is differentiable, then the Implicit Function Theorem enables us to compute $x'(t)$. By pursuing this idea, we can describe the curve in (5) by a differential equation. Assuming an arbitrary homotopy, we have

$$0 = h(t, x(t)) \quad (6)$$

On differentiating with respect to t , we obtain

$$0 = h_t(t, x(t)) + h_x(t, x(t))x'(t) \quad (7)$$

in which subscripts denote partial derivatives. Thus,

$$x'(t) = -[h_x(t, x(t))]^{-1} h_t(t, x(t)) \quad (8)$$

This is a differential equation for x . It has a known initial value because $x(0)$ is supposedly known. On integrating this differential equation (usually by numerical procedures), we shall have the value $x(1)$, which is the solution.

EXAMPLE 1 We let $X = Y = \mathbb{R}^2$, and define

$$f(x) = \begin{bmatrix} \xi_1^2 - 3\xi_2^2 + 3 \\ \xi_1\xi_2 + 6 \end{bmatrix} \quad x = (\xi_1, \xi_2) \in \mathbb{R}^2$$

Solution A convenient homotopy is given by Equation (4), and we select $x_0 = (1, 1)$. The derivatives on the right side of Equation (8) are computed to be

$$h_x = f'(x) = \begin{bmatrix} \partial f_1 / \partial \xi_1 & \partial f_1 / \partial \xi_2 \\ \partial f_2 / \partial \xi_1 & \partial f_2 / \partial \xi_2 \end{bmatrix} = \begin{bmatrix} 2\xi_1 & -6\xi_2 \\ \xi_2 & \xi_1 \end{bmatrix}$$

$$h_t = f(x_0) = \begin{bmatrix} f_1(x_0) \\ f_2(x_0) \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

The inverse of $f'(x)$ is

$$h_x^{-1} = [f'(x)]^{-1} = \frac{1}{\Delta} \begin{bmatrix} \xi_1 & 6\xi_2 \\ -\xi_2 & 2\xi_1 \end{bmatrix} \quad \Delta = 2\xi_1^2 + 6\xi_2^2$$

The differential equation that governs the path leading away from the point x_0 is Equation (8). In this concrete case, it is a pair of ordinary differential equations:

$$\begin{bmatrix} \xi_1' \\ \xi_2' \end{bmatrix} = -\frac{1}{\Delta} \begin{bmatrix} \xi_1 & 6\xi_2 \\ -\xi_2 & 2\xi_1 \end{bmatrix} \begin{bmatrix} 1 \\ 7 \end{bmatrix} = -\frac{1}{\Delta} \begin{bmatrix} \xi_1 + 42\xi_2 \\ -\xi_2 + 14\xi_1 \end{bmatrix}$$

This system was integrated numerically (using one of the methods to be presented in Chapter 8, p. 524) over the interval $0 \leq t \leq 1$, and at $t = 1$ the solution was $(-2.961, 1.978)$. Notice that f has a root $(-3, 2)$.

To complete this problem and find its numerical solution, we can use Newton's iteration starting at the point produced by the homotopy method. The Newton iteration replaces any approximate root x by $x - \delta$, where the correction δ is given by

$$\delta = [f'(x)]^{-1} f(x)$$

In this example, the vector δ is

$$\begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix} = \frac{1}{\Delta} \begin{bmatrix} \xi_1 & 6\xi_2 \\ -\xi_2 & 2\xi_1 \end{bmatrix} \begin{bmatrix} \xi_1^2 - 3\xi_2^2 + 3 \\ \xi_1\xi_2 + 6 \end{bmatrix}$$

Three steps of the Newton iteration produced these results:

k	ξ_1	ξ_2
0	-2.96100 00000 00	1.97800 00000 00
1	-3.00025 32813 14	2.00032 02744 78
2	-3.00000 00057 80	2.00000 00378 24
3	-3.00000 00000 00	2.00000 00000 00

The following formal result, due to Ortega and Rheinboldt [1970], gives some conditions under which the homotopy method will succeed.

THEOREM 1 Theorem on Continuously Differentiable Solution

If $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuously differentiable and if $\| [f'(x)]^{-1} \| \leq M$ on $\mathbb{R}^n$, then for any $x_0 \in \mathbb{R}^n$ there is a unique curve $\{x(t) : 0 \leq t \leq 1\}$ in $\mathbb{R}^n$ such that $f(x(t)) + (t - 1)f(x_0) = 0$, with $0 \leq t \leq 1$. The function $t \mapsto x(t)$ is a continuously differentiable solution of the initial-value problem $x' = -[f'(x)]^{-1} f(x_0)$, where $x(0) = x_0$.

Tracing the Path

Another way of tracing the path $x(t)$ has been described by Garcia and Zangwill [1981]. We start with the equation $h(t, x) = 0$, supposing that $x \in \mathbb{R}^n$ and $t \in [0, 1]$. A vector $y \in \mathbb{R}^{n+1}$ is defined by

$$y = (t, \xi_1, \xi_2, \dots, \xi_n)$$

where $\xi_1, \xi_2, \dots, \xi_n$ are the components of x . Hence, our equation is simply $h(y) = 0$. Each component of y , including t , is now allowed to be a function of an independent variable s , and we write $h(y(s)) = 0$. Differentiating with respect to s , we obtain the basic differential equation

$$h'(y(s))y'(s) = 0 \tag{9}$$

The variable s starts at 0, as does t . The initial value of x is $x(0) = x_0$. Thus, suitable starting values are available for the differential equation (9).

Since f and g are maps of $\mathbb{R}^n$ into $\mathbb{R}^n$, h is a map of $\mathbb{R}^{n+1}$ into $\mathbb{R}^n$. The derivative $h'(y)$ is therefore represented by an $n \times (n + 1)$ matrix, A . The vector $y(s)$ has $n + 1$ components, which we denote by $\eta_1, \eta_2, \dots, \eta_{n+1}$. By appealing to the lemma below, we can obtain another form for Equation (9); namely,

$$\eta'_j = (-1)^{j+1} \det(A_j) \quad (1 \leq j \leq n + 1) \tag{10}$$

where A_j is the $n \times n$ matrix that results from A by deleting the j th column. Let us illustrate this formalism with the same problem as in Example 1.

EXAMPLE 2 Taking f and x_0 as in Example 1, we have

$$h(t, x) = \begin{bmatrix} \xi_1^2 - 3\xi_2^2 + 2 + t \\ \xi_1\xi_2 - 1 + 7t \end{bmatrix}$$

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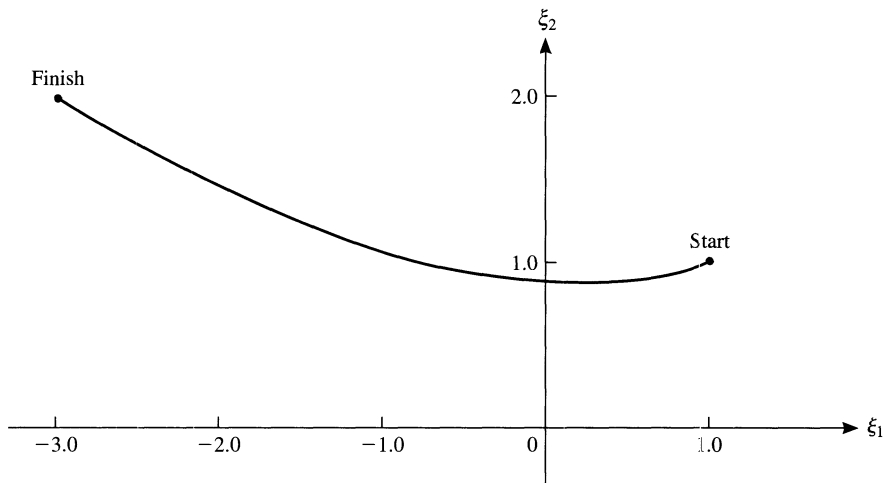


FIGURE 3.9
The path generated in
Example 2

Solution The differential equation (9) is given by

$$\begin{bmatrix} \partial h_1 / \partial t & \partial h_1 / \partial \xi_1 & \partial h_1 / \partial \xi_2 \\ \partial h_2 / \partial t & \partial h_2 / \partial \xi_1 & \partial h_2 / \partial \xi_2 \end{bmatrix} \begin{bmatrix} t' \\ \xi_1' \\ \xi_2' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
$$\begin{bmatrix} 1 & 2\xi_1 & -6\xi_2 \\ 7 & \xi_2 & \xi_1 \end{bmatrix} \begin{bmatrix} t' \\ \xi_1' \\ \xi_2' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{11}$$

It is preferable to use Equation (10), however, and to write the differential equations in the form

$$\begin{cases} t' = 2\xi_1^2 + 6\xi_2^2 & t(0) = 0 \\ \xi_1' = -\xi_1 - 42\xi_2 & \xi_1(0) = 1 \\ \xi_2' = \xi_2 - 14\xi_1 & \xi_2(0) = 1 \end{cases} \tag{12}$$

The derivatives in this system are with respect to s . Performing a numerical integration, we arrive at these two points:

$$\begin{array}{llll} s = 0.087 & t = 0.969 & \xi_1 = -2.94 & \xi_2 = 1.97 \\ s = 0.088 & t = 1.010 & \xi_1 = -3.02 & \xi_2 = 2.01 \end{array}$$

Either of these can be used to start a Newton iteration, as was done in Example 1. The path generated by this homotopy is shown in Figure 3.9. ■

A drawback to the method used in Example 2 is that we have no *a priori* knowledge of the value of s corresponding to $t = 1$. In practice, this may necessitate several computer runs.

LEMMA 1 Lemma on Solution of Homogeneous Equation

Let A be an $n \times (n + 1)$ matrix. A solution of the homogeneous equation $Ax = 0$ is given by $x_j = (-1)^j \det(A_j)$, where A_j is A without column j .

Proof Select any row (for example, the i th row) in A , and adjoin a copy of it as a new row at the top of A . This creates an $(n + 1) \times (n + 1)$ matrix B that is obviously singular because row i of A occurs twice in B . In expanding the determinant of B by the elements in its top row, we obtain

$$0 = \det B = \sum_{j=1}^{n+1} (-1)^{j+1} a_{ij} \det(A_j) = - \sum_{j=1}^{n+1} a_{ij} x_j$$

Since this is true for $i = 1, 2, \dots, n$, we have $Ax = 0$. ■

Relation to Newton's Method

The connection between the homotopy methods and Newton's method is deeper than may be seen at first glance. Let us start with the homotopy

$$h(t, x) = f(x) - e^{-t} f(x_0) \quad (13)$$

In this equation, t will run from 0 to ∞ . We seek a curve or path, $x = x(t)$, on which

$$0 = h(t, x(t)) = f(x(t)) - e^{-t} f(x_0)$$

As usual, differentiation with respect to t will lead to a differential equation describing the path:

$$\begin{aligned} 0 &= f'(x(t))x'(t) + e^{-t} f(x_0) \\ &= f'(x(t))x'(t) + f(x(t)) \\ x'(t) &= -[f'(x(t))]^{-1} f(x(t)) \end{aligned} \quad (14)$$

If this differential equation is integrated with Euler's method (described in Section 8.2, p. 534), with step size 1, the result is the formula

$$x_{n+1} = x_n - [f'(x_n)]^{-1} f(x_n)$$

This is, of course, the formula for Newton's method. It is clear that one can expect to obtain better results by integrating Equation (14) with a more accurate numerical method and a variable step size. (For these matters, consult Chapter 8, p. 524.)

Linear Programming

The homotopy method can be used to solve linear programming problems. (Such problems are discussed in Section 10.3, p. 695.) This approach leads naturally to the algorithm proposed by Karmarkar [1984]. In explaining the homotopy method in this context, we follow closely the description given by Brophy and Smith [1988].

The reader who wishes to explore these ideas further can profitably consult other references.

Consider the standard linear programming problem

$$\begin{cases} \text{maximize } c^T x \\ \text{subject to } Ax = b \text{ and } x \geq 0 \end{cases} \tag{15}$$

Here, $c \in \mathbb{R}^n$, $x \in \mathbb{R}^n$, $b \in \mathbb{R}^m$, and A is an $m \times n$ matrix. We start with a **feasible point**—that is, a point x^0 that satisfies the constraints. The **feasible set** is

$$\mathcal{F} = \{x \in \mathbb{R}^n : Ax = b \text{ and } x \geq 0\}$$

Our intention is to move from $x^{(0)}$ to a succession of other points, remaining always in $\mathcal{F}$, and increasing the value of the **objective function**, $c^T x$. It is clear that if we move from $x^{(0)}$ to $x^{(1)}$, the difference $x^{(1)} - x^{(0)}$ must lie in the null space of A . We shall try to find a curve $t \mapsto x(t)$ in the feasible set, starting at x^0 and leading to a solution of the extremal problem. Our requirements are

- 1. $x(t) \geq 0$ for $t \geq 0$
- 2. $Ax(t) = b$ for $t \geq 0$
- 3. $c^T x(t)$ is increasing for $t \geq 0$.

The curve will be defined by an initial-value problem:

$$\begin{cases} x' = f(x) \\ x(0) = x^{(0)} \end{cases} \tag{16}$$

The task facing us is to determine a suitable f . To satisfy Condition 1, we shall arrange that whenever a component x_i approaches 0, its velocity $x'_i(t)$ will also approach 0. This can be accomplished by putting

$$D(x) = \begin{bmatrix} x_1 & & & 0 \\ & x_2 & & \\ & & \ddots & \\ 0 & & & x_n \end{bmatrix}$$

and assuming that for some bounded function G ,

$$f(x) = D(x)G(x) \tag{17}$$

If this is the case, then from Equations (16) and (17) we have

$$x'_i = x_i G_i(x)$$

and clearly $x'_i \rightarrow 0$ if $x_i \rightarrow 0$.

To satisfy Requirement 2, it suffices to require $Ax' = 0$. Since $x' = f = DG$, we must require $ADG = 0$. This is most conveniently arranged by letting $G = PH$ where H is any function and P is the orthogonal projection onto the null space of AD .

Finally, to satisfy Property 3, we should select H so that $c^T x(t)$ is increasing. Thus, we want

$$0 < \frac{d}{dt}(c^T x(t)) = c^T x' = c^T f(x) = c^T DG = c^T DPH$$

A convenient choice for H is Dc because then we have, with $v = Dc$,

$$\begin{aligned} c^T DPH &= c^T DP Dc = v^T P v = \langle v, P v \rangle \\ &= \langle v - P v + P v, P v \rangle = \langle P v, P v \rangle \geq 0 \end{aligned}$$

Notice that $v - P v$ is orthogonal to the range of P , and $\langle v - P v, P v \rangle = 0$.

The final version of our initial-value problem is

$$x' = D(x)P(x)D(x)c \quad x(0) = x^{(0)} \quad (18)$$

The theoretical formula for P is

$$P = I - (AD)^T [(AD)(AD)^T]^{-1} AD \quad (19)$$

The validity of this depends on $B \equiv AD$ having full rank, so that BB^T will be nonsingular. This will, in turn, require $x_i > 0$ for each component. Thus, the points $x(t)$ should remain in the interior of the set

$$\{x : x \geq 0\}$$

In particular, $x^{(0)}$ should be so chosen. In practice, $P v$ is computed not by Equation (19) but by solving the equation $BB^T z = B v$ and noting that

$$P v = v - B^T z$$

The initial-value problem (18) need not be solved very accurately. A variation of the Euler method can be used. Recall that the Euler method for Equation (16) advances the solution by

$$x(t + \delta) = x(t) + \delta x'(t) = x(t) + \delta f(x)$$

Using this type of formula, we generate a sequence of vectors $x^{(0)}, x^{(1)}, \dots$ by the equation

$$x^{(k+1)} = x^{(k)} + \delta_k f(x^{(k)})$$

Although it is tempting to take the value of δ_k as large as possible subject to the requirement $x^{(k+1)} \in \mathcal{F}$, that will lead to a point $x^{(k+1)}$ having at least one 0 component. As mentioned previously, that will introduce other difficulties. What seems to work well in practice is to take δ_k approximately 9/10 of the maximum possible step. The latter is easily computed; it is the maximum λ for which $x^{(k+1)} \geq 0$. (The constraint $Ax = b$ is maintained automatically.)

PROBLEMS 3.6

1. Solve the system of equations

$$x - 2y + y^2 + y^3 - 4 = -x - y + 2y^2 - 1 = 0$$

by the homotopy method used in Example 2, starting with the point $(0, 0)$. (All the calculations can be performed without recourse to numerical methods.)

2. Consider the homotopy $h(t, x) = tf(x) + (1 - t)g(x)$, in which

$$f(x) = x^2 - 5x + 6 \quad g(x) = x^2 - 1$$

Show that there is no path connecting a root of g to a root of f .

3. Let $y = y(s)$ be a differentiable function from $\mathbb{R}$ to $\mathbb{R}^n$ satisfying the differential equation (9). Assume that $h(y(0)) = 0$. Prove that $h(y(s)) = 0$.
4. If the homotopy method of Example 2 is to be used on the system

$$\sin x + \cos y + e^{xy} = \tan^{-1}(x + y) - xy = 0$$

starting at $(0, 0)$, what system of differential equations will govern the path? A computer program to seek the solution will be instructive.

5. Prove that homotopy is an equivalence relation among the continuous maps from one topological space to another.
6. Are the functions $f(x) = \sin x$ and $g(x) = \cos x$ homotopic?
7. Consider these maps of $[0, 1]$ into $[0, 1] \cup [2, 3]$:

$$f(x) = 0 \quad g(x) = 2$$

Are they homotopic?